西南交通大学
Southwest Jiaotong University

SCAI011112 视觉计算

表征学习

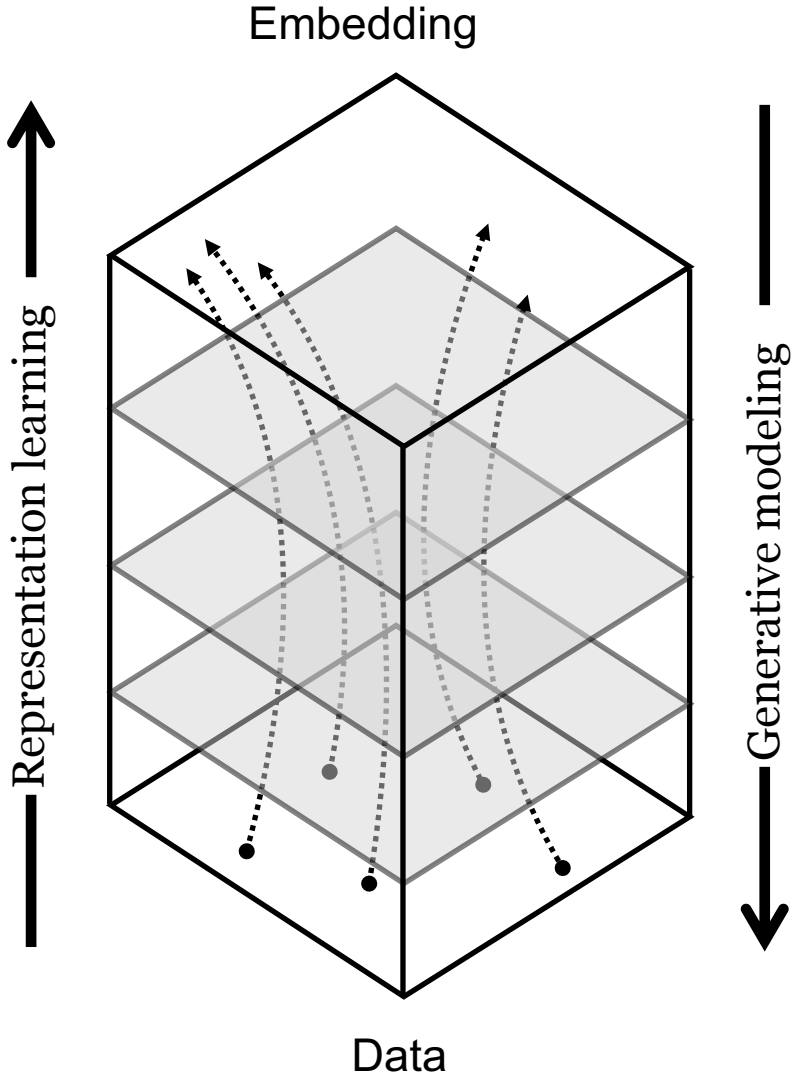
计算机科学与技术专业
2025-2026 (2)

本章主要内容

- 表征学习问题
- 基于**压缩**的学习原则
  - 自编码器，对比学习，信息瓶颈，聚类
- 基于**预测**的学习原则
  - 预测编码，自监督学习，插补
- 信息论与**信息瓶颈**

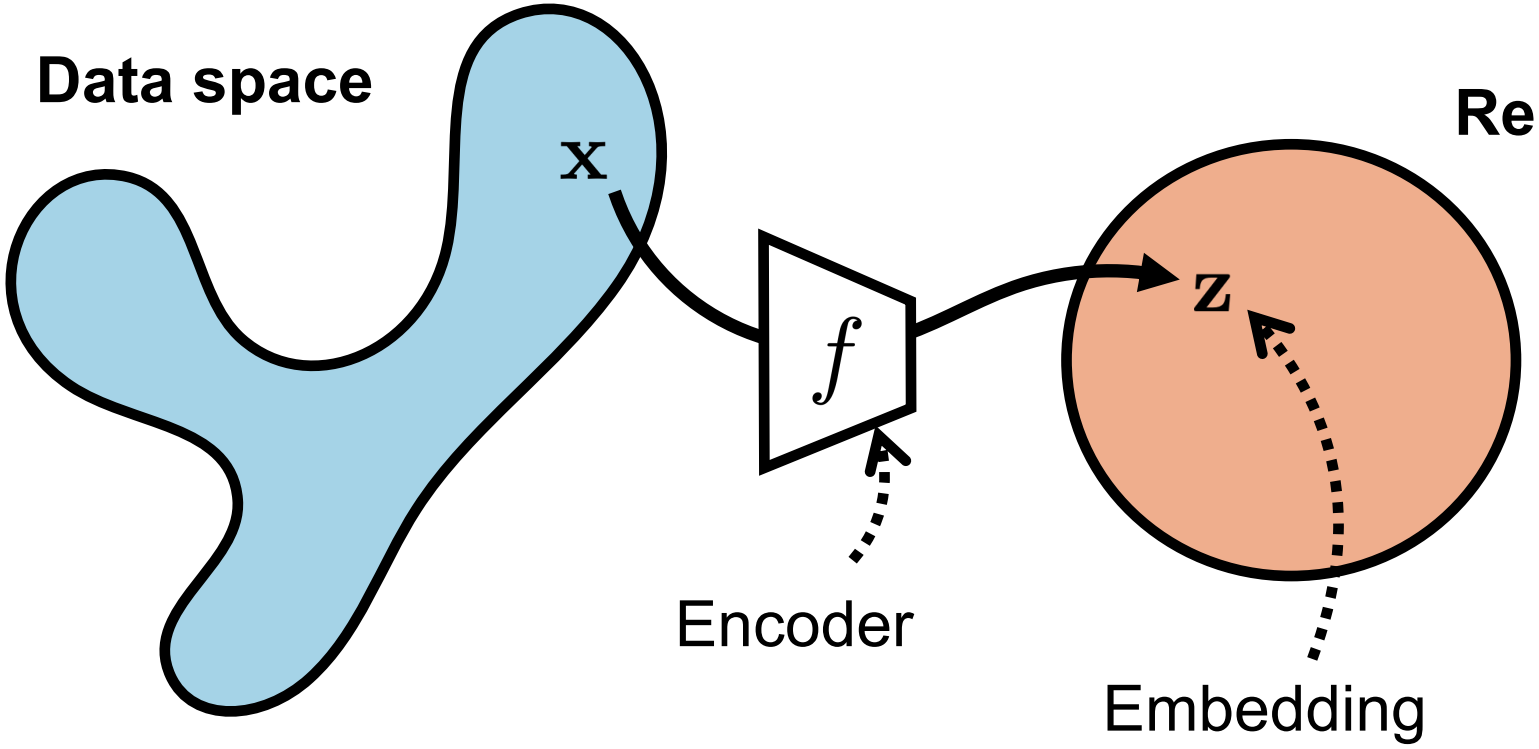
表征学习问题

- Deep nets transform datapoints, layer by layer
- Each layer is a different representation of the data
- In the forward direction, the mapping goes from observed data to latent embeddings — this direction is called **representation learning**
- In the reverse direction, the mapping goes from latent embeddings to observed data — this direction is called **generative modeling**



表征学习问题

- 表示学习的目标是学习如何将数据点 $\mathbf{x} \in \mathcal{X}$ 映射到抽象表示 $\mathbf{z} \in \mathcal{Z}$
- 将这种映射称为“编码”，并学习一个函数 $f: \mathcal{X} \to \mathcal{Z}$。通常 $\mathbf{x}$ 和 $\mathbf{z}$都是高维向量，而 $\mathbf{z}$被称为 $\mathbf{x}$的**向量嵌入**



本章我们关注目标函数，而不是网络结构

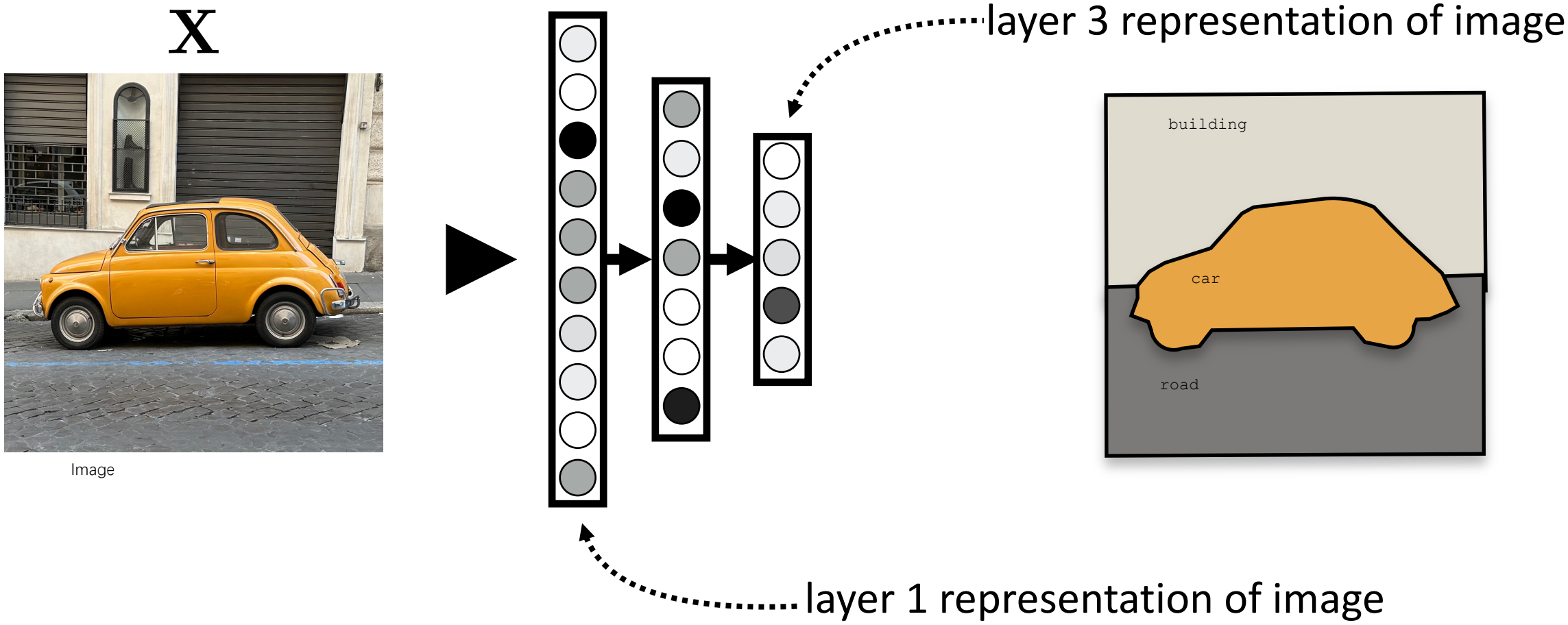
表征学习问题

训练映射 f 的目的是让 $\mathbf{z}$ 具备某些属性：

- $\mathbf{z}$ 的维度低于 $\mathbf{x}$
- $\mathbf{z}$ 值的分布 $p(\mathbf{z})$ 具有简单的结构（例如单位正态分布）
- $\mathbf{z}$ 各个维度是解耦的（disentangled）
- $\mathbf{z}$ 是可解释的
- ……

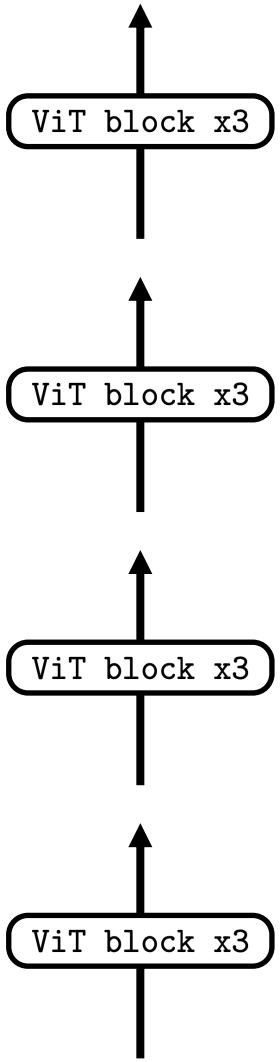
z成为一个比 x 更简单、更抽象或组织得更好的表示空间

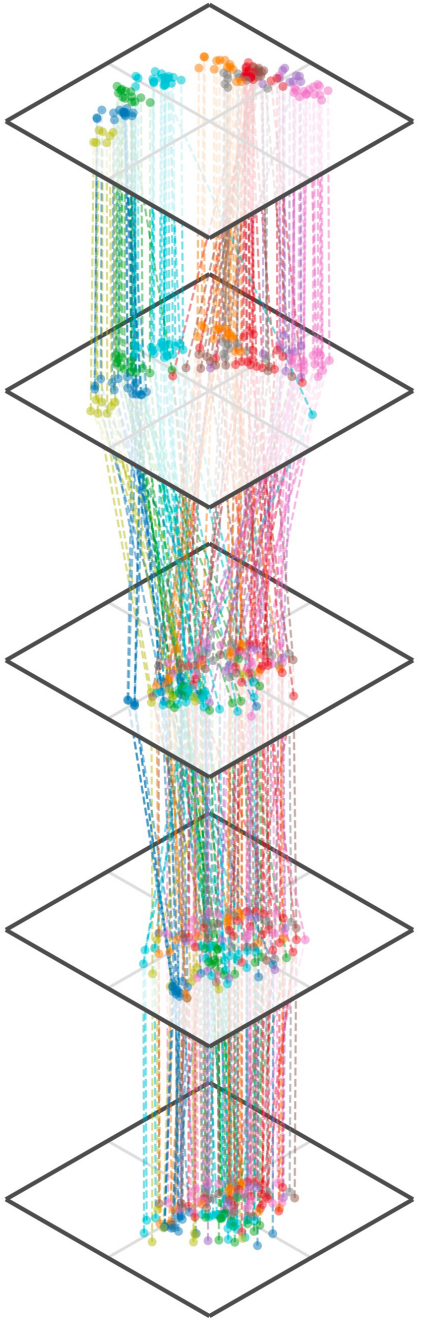
表征学习问题

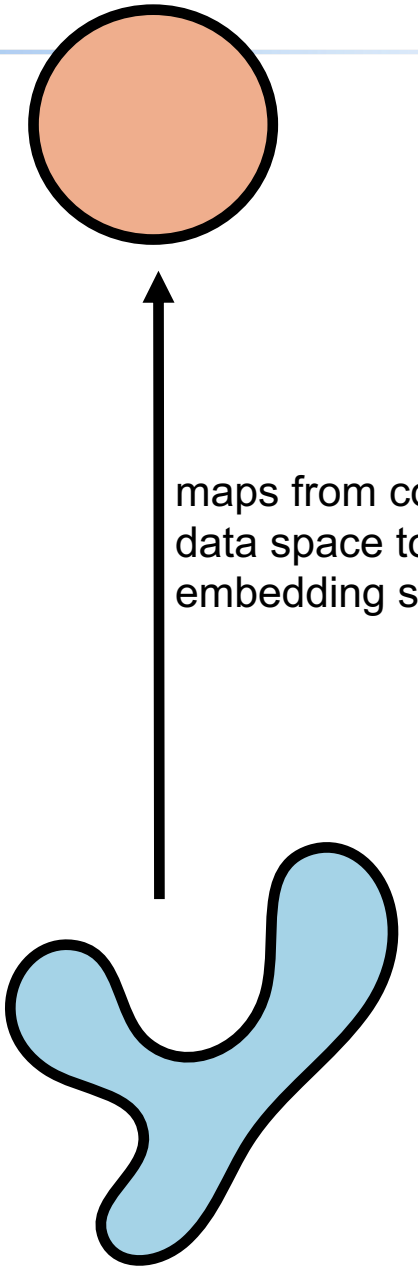


Represent data as a neural embedding — a vector/tensor of neural activations (perhaps representing a vector of detected texture patterns or object parts)

CLIP







[Radford*, Kim* et al., ICML 2021]

压缩

好的表示是简约的（parsimonious），并且仅捕捉数据中对于人类感兴趣的任务所必需的本质特征。压缩表示的必要性至少体现在以下三个层面：

- 压缩的表示需要更少的存储内存
- 压缩是实现对无关干扰因素（nuisance factors）保持不变性的一种方式
- 压缩是奥卡姆剃刀法则（Occam's razor）的一种体现：在所有能同等很好地解释数据的竞争假设中，最简单的往往最可能是真实的
  - 在同样能很好地拟合数据的两种表示中，优先选择压缩程度更高的那一个。

压缩的本质是“剥离伪相关”

预测

- 拥有视觉系统的全部意义在于能够采取行动以实现理想的未来结果。
- 一般来说，预测可以指在给定观测数据的情况下，对世界任意属性的推断，包括未来、过去、心理状态、因果关系等。要做这些预测，好的表示具有决定性作用
- 视觉领域的大多数表示学习算法都是关于学习世界的压缩编码，这些编码同时也要对未来具有预测能力

表征学习器的类型

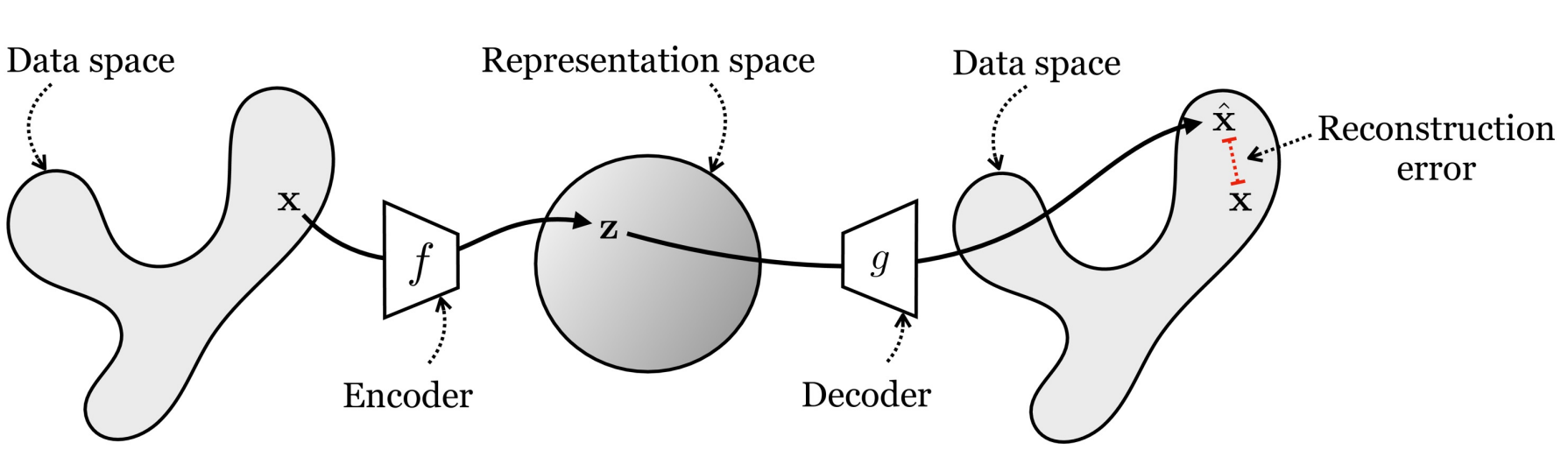
学习方法 (Learning Method)	学习原则 (Learning Principle)	简短摘要 (Short Summary)
自编码 (Autoencoding)	压缩 (Compression)	移除冗余信息
对比学习 (Contrastive)	压缩 (Compression)	实现对观察变换的不变性
聚类 (Clustering)	压缩 (Compression)	将连续数据量化为离散类别
未来预测 (Future prediction)	预测 (Prediction)	预测未来
缺失填补 (Imputation)	预测 (Prediction)	预测缺失数据
代理任务 (Pretext tasks)	预测 (Prediction)	预测数据的抽象属性

本章主要内容

- 表征学习问题
- 基于**压缩**的学习原则
  - 自编码器，对比学习，信息瓶颈，聚类
- 基于**预测**的学习原则
  - 预测编码，自监督学习，插补
- 信息论与**信息瓶颈**

自编码器（Autoencoders）

- 自编码器是一个将数据映射回其自身的函数，但必须经过一个低维的表示瓶颈



$$f^*, g^* = \arg\min_{f,g} \mathbb{E}_{\mathbf{x}} \|g(f(\mathbf{x})) - \mathbf{x}\|_2^2$$

自编码器（Autoencoders）

Data

$\{\mathbf{x}^{(i)}\}_{i=1}^{n} \rightarrow$

L_2 autoencoder

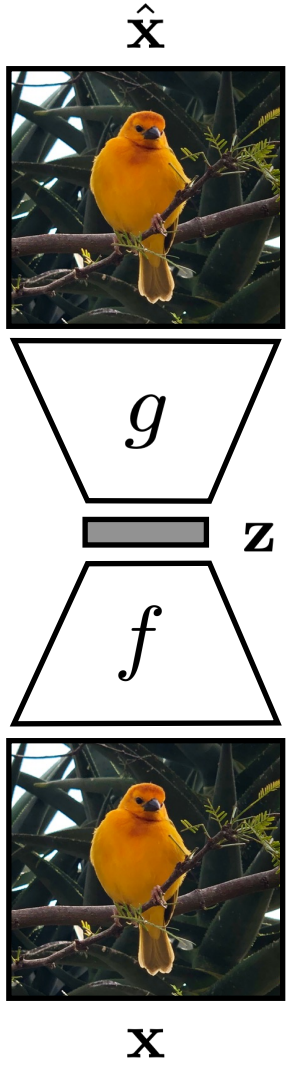
Objective

$$\mathcal{L}(F(\mathbf{x}), \mathbf{x}) = \|F(\mathbf{x}) - \mathbf{x}\|_2^2$$

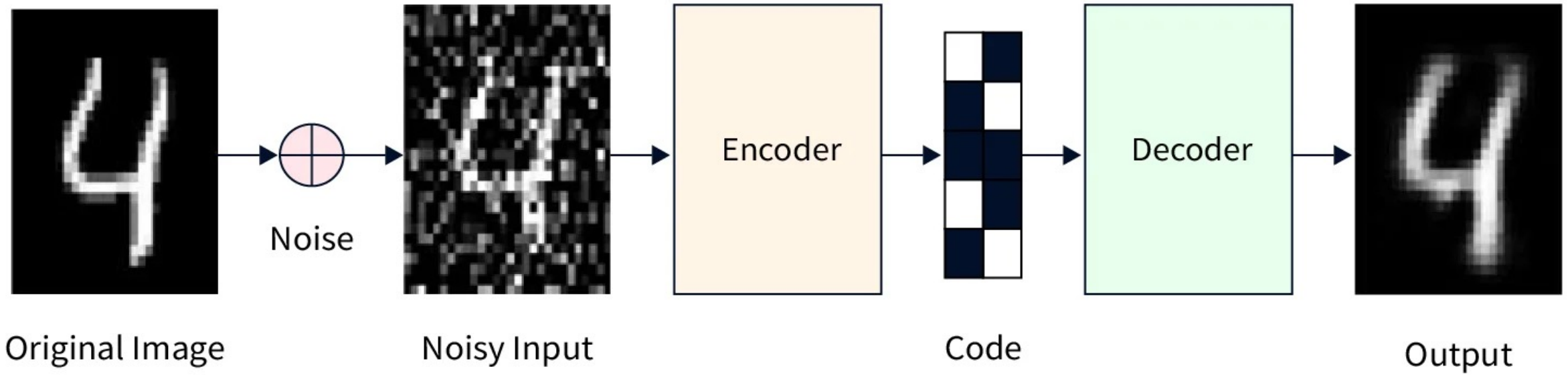
Hypothesis space

$$F = g \circ f : \mathbb{R}^N \rightarrow \mathbb{R}^M \rightarrow \mathbb{R}^N$$

$\rightarrow f$

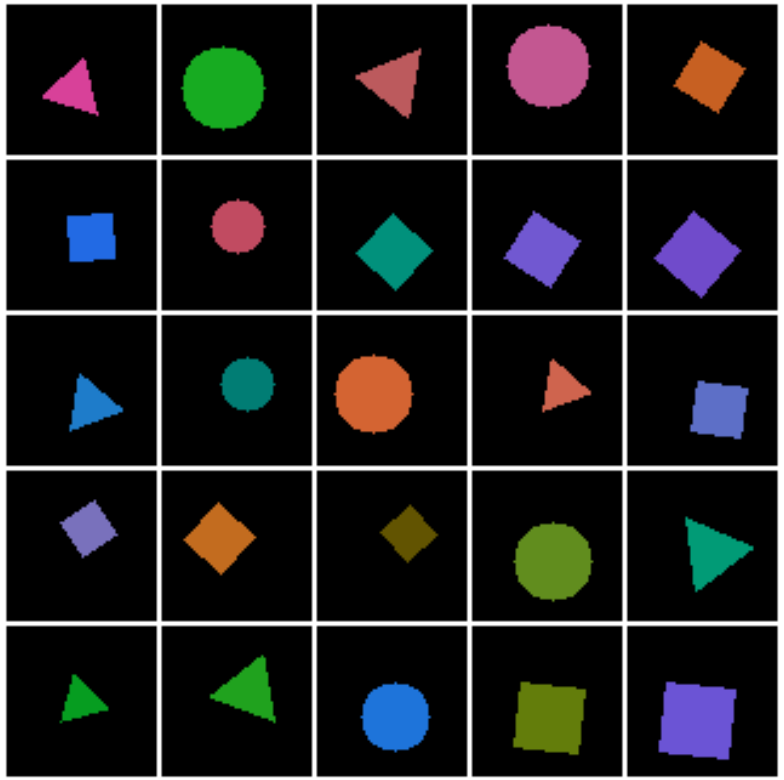


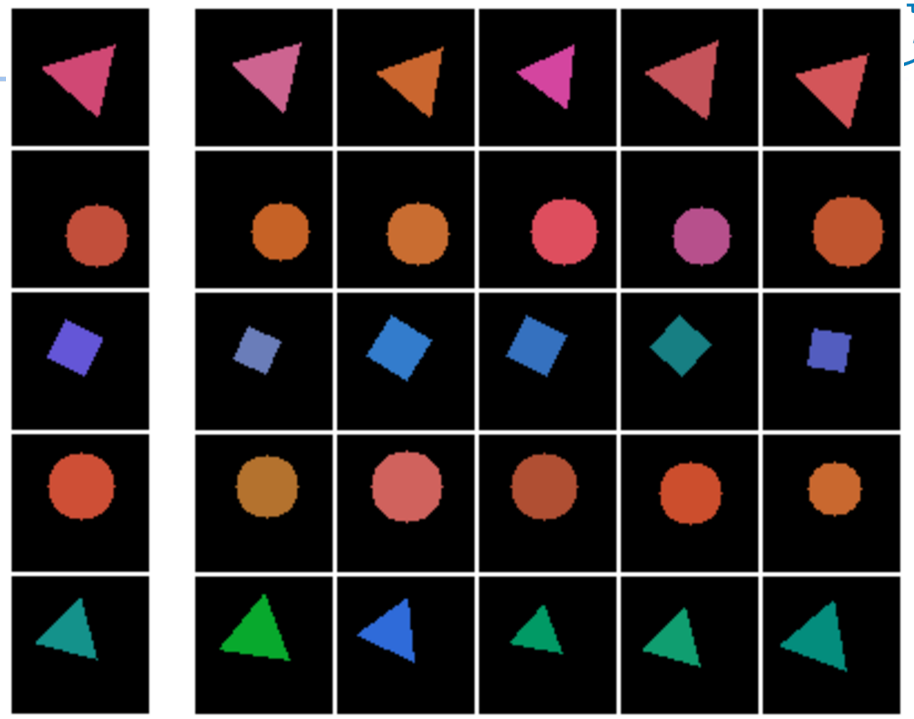
去噪自编码器（Denoising Autoencoders）

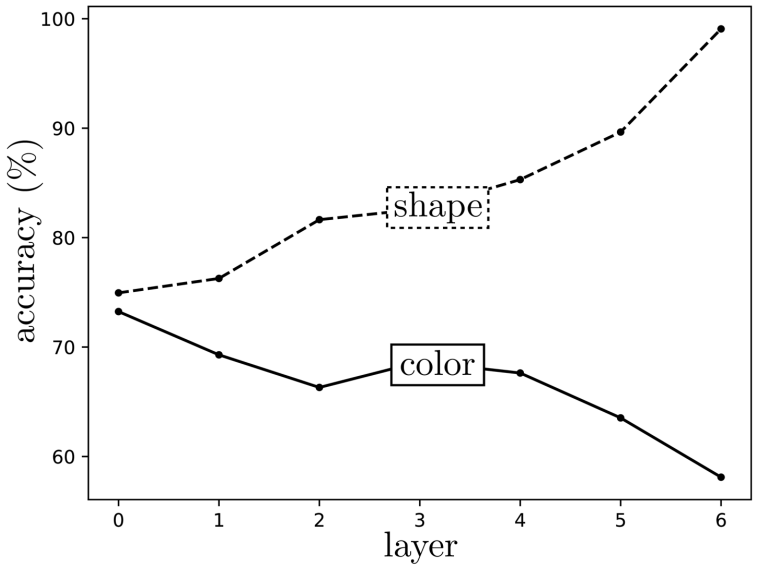


自编码器（Autoencoders）

- Dataset consists just of colored circles, triangles, and squares.
- Each shape has a randomized size, position, and rotation. Each shape's color is sampled from one of eight color classes (orange, green, purple, etc.) plus a small random perturbation.

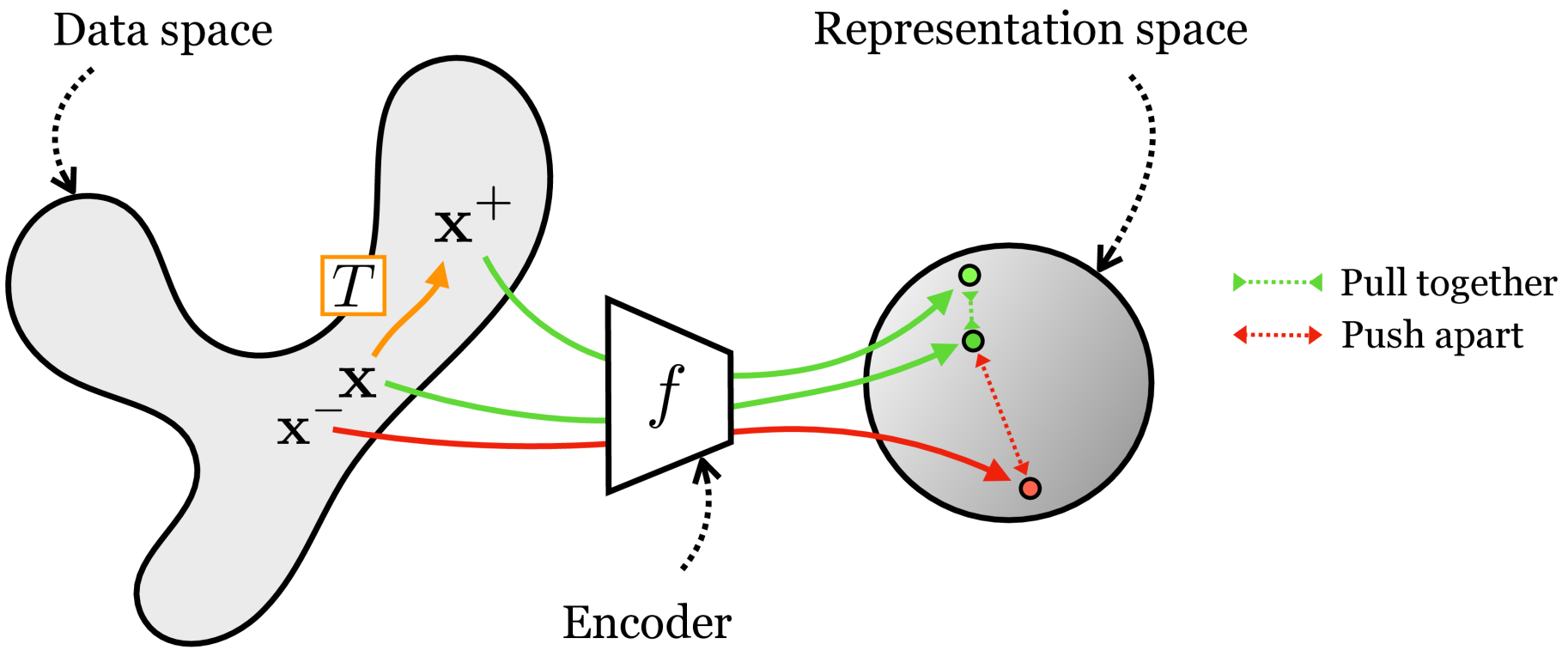




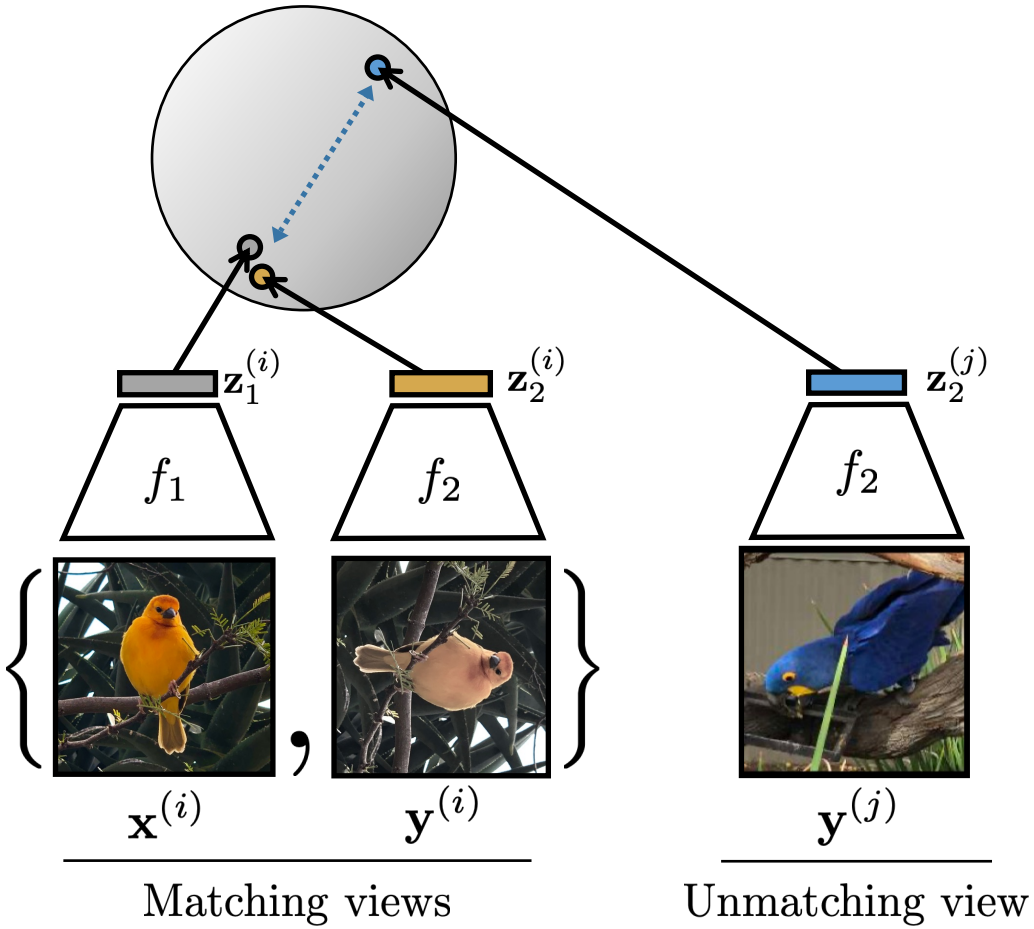


对比学习

- 在对比学习中，表征被监督对特定的观察变换保持“**不变性**”，从而产生一种仅捕获不同数据“视图”之间共有属性的压缩表示
  - 监督信号以正样本对和负样本对的形式提供



对比学习 — 基于变换



Data

$\{\mathbf{x}^{(i)}\}_{i=1}^{N}, T \longrightarrow$

Constrastive learning (transformations)

Objective

$\sum_{i,j} D(f(T(\mathbf{x}^{(i)})), f(\mathbf{x}^{(i)}))$

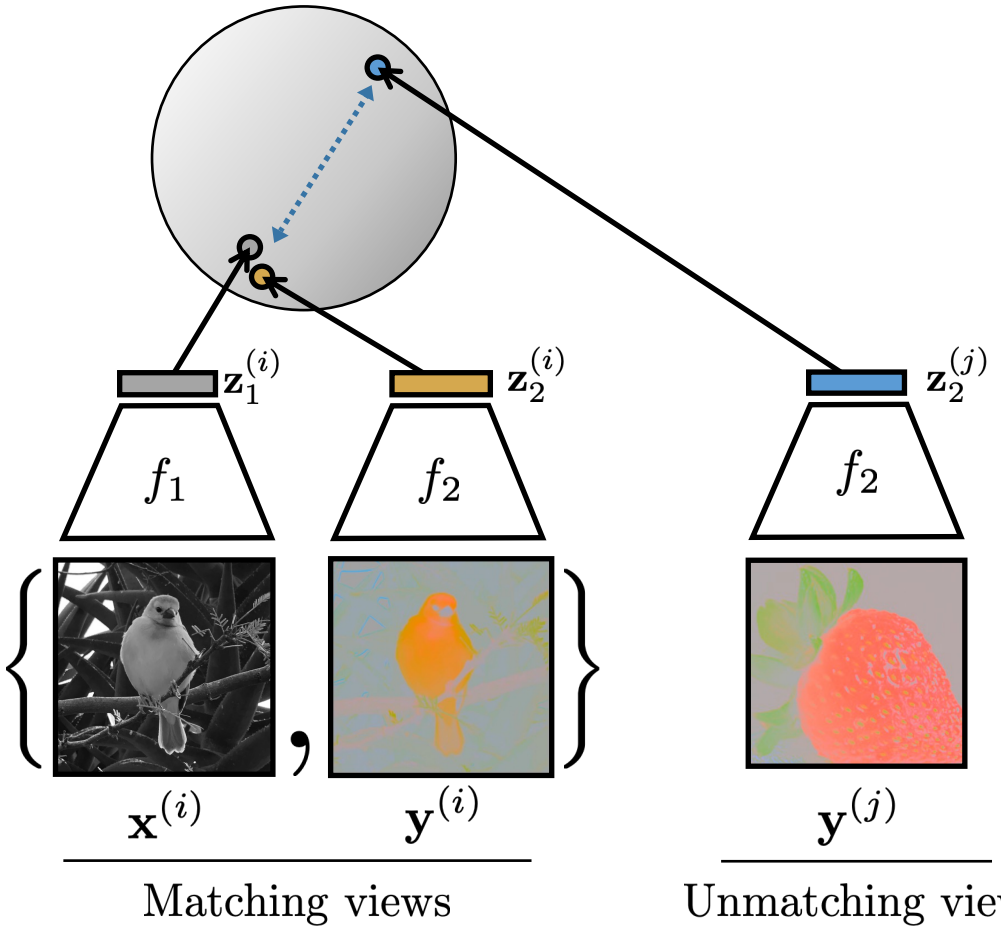
$-D(f(\mathbf{x}^{(i)}), f(\mathbf{x}^{(j)}))$

Hypothesis space

$f : \mathbb{R}^N \to \mathbb{R}^M$

$\longrightarrow f$

对比学习 — 基于共现



Data

$\{\mathbf{x}^{(i)}, \mathbf{y}^{(i)}\}_{i=1}^{N} \rightarrow$

Constrastive learning (co-occurrences)

Objective

$$\sum_{i,j} D(f_1(\mathbf{x}^{(i)}), f_2(\mathbf{y}^{(i)}))$$

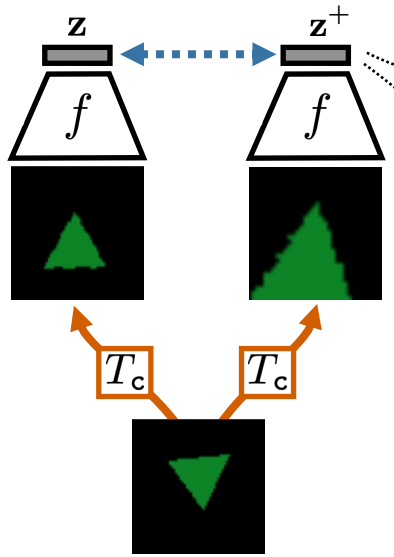
$$-D(f_1(\mathbf{x}^{(i)}), f_2(\mathbf{y}^{(j)}))$$

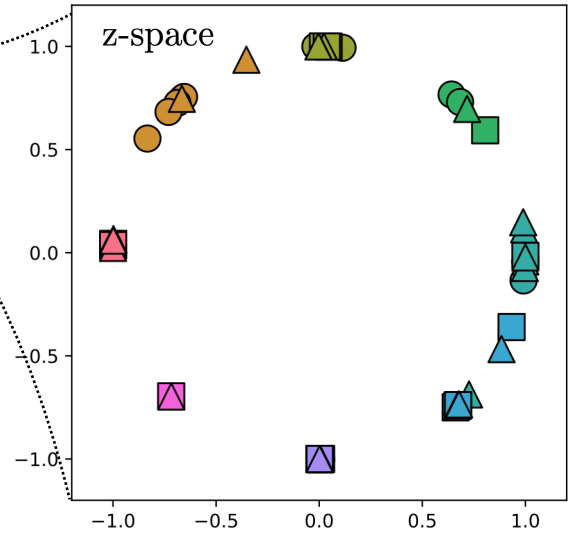
Hypothesis space

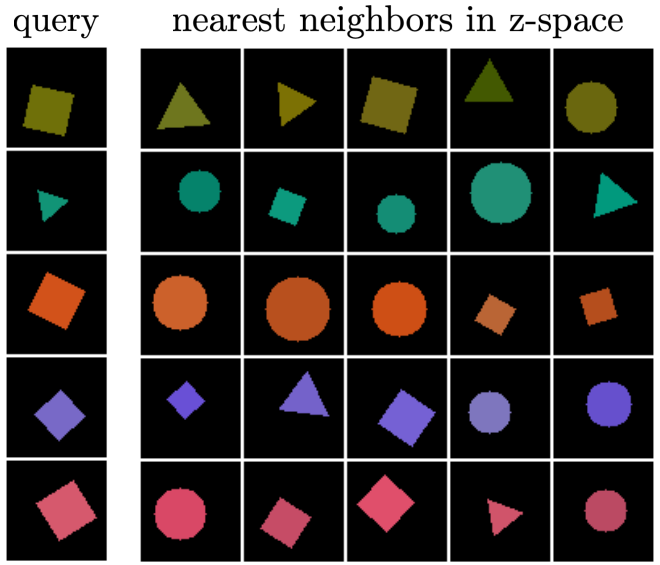
$$f_1, f_2 : \mathbb{R}^N \rightarrow \mathbb{R}^M$$

$\rightarrow f_1, f_2$

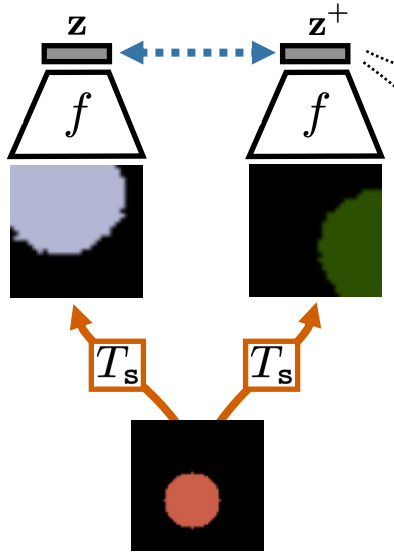
Color sensitive contrastive learning

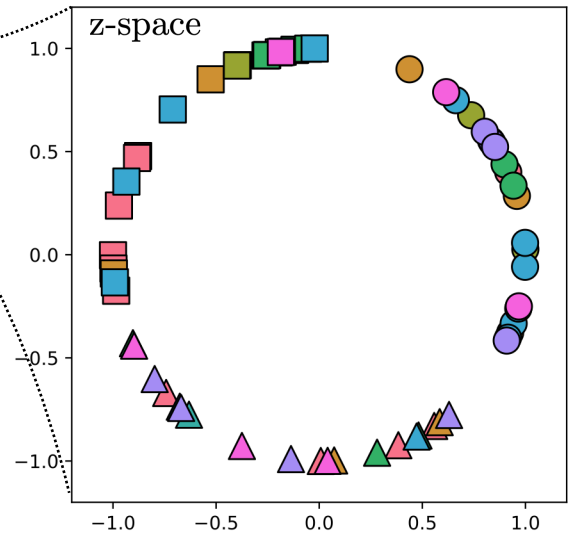


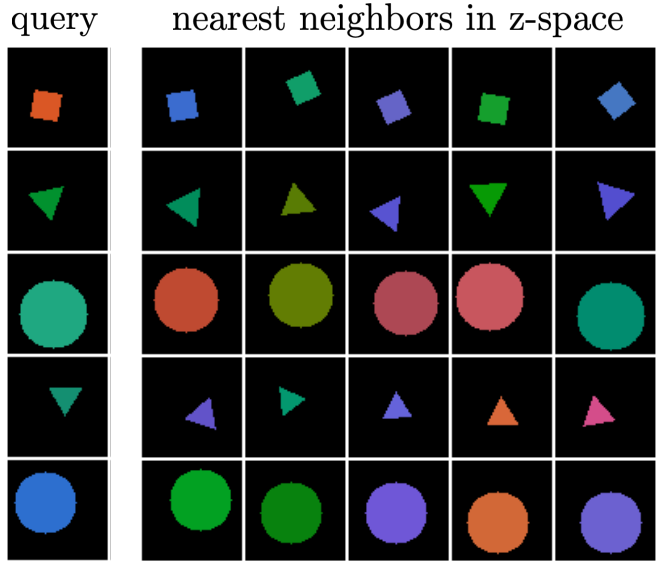




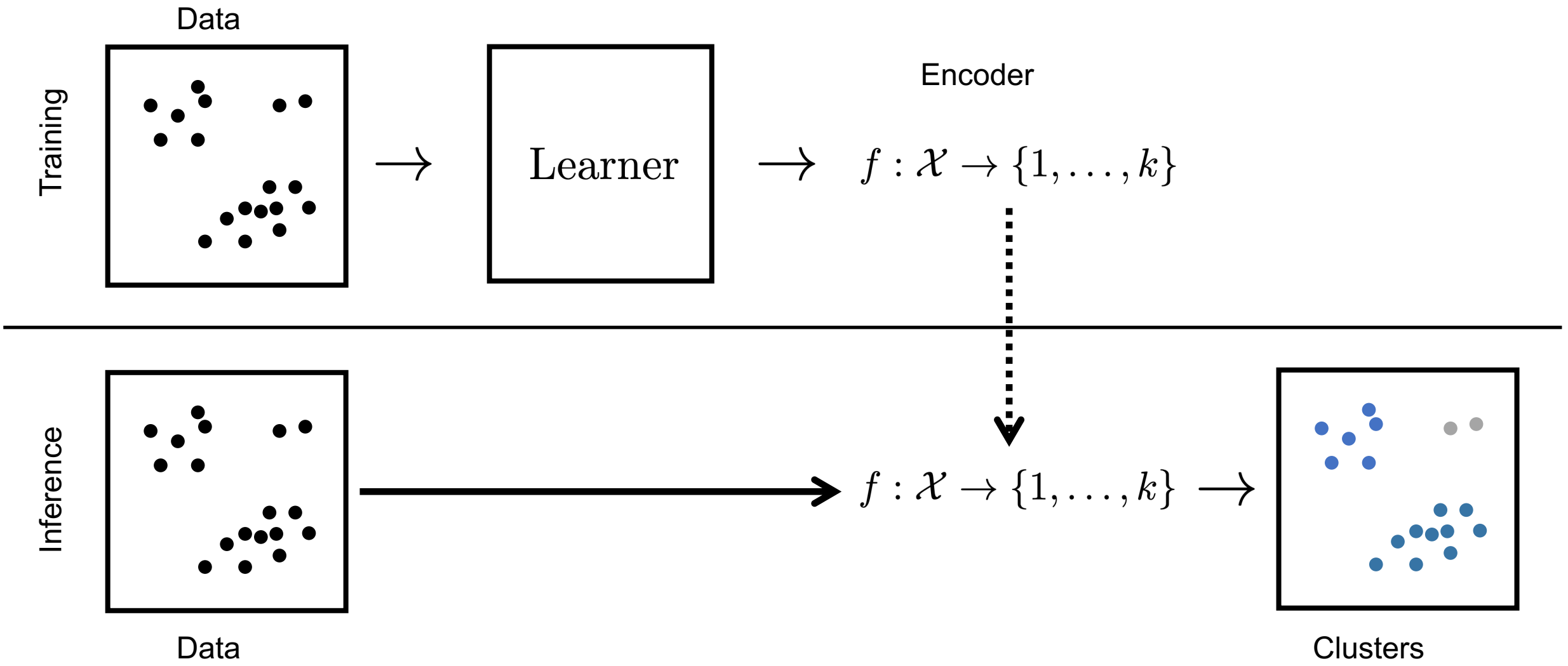
Shape sensitive contrastive learning



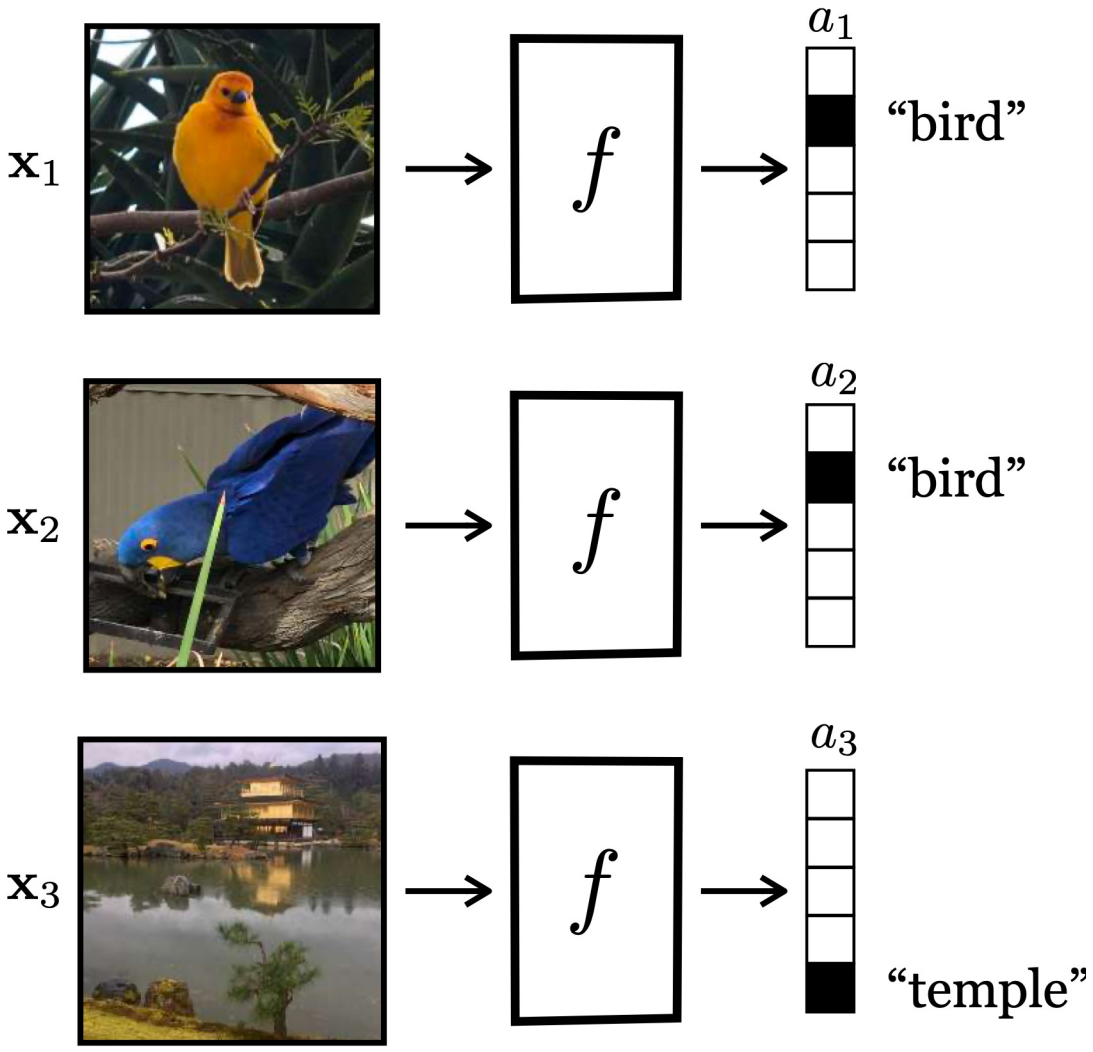




聚类


Data
Training
Learner
Encoder
$f : \mathcal{X} \to \{1, \ldots, k\}$
Inference
$f : \mathcal{X} \to \{1, \ldots, k\}$
Data
Clusters

基于聚类的表征

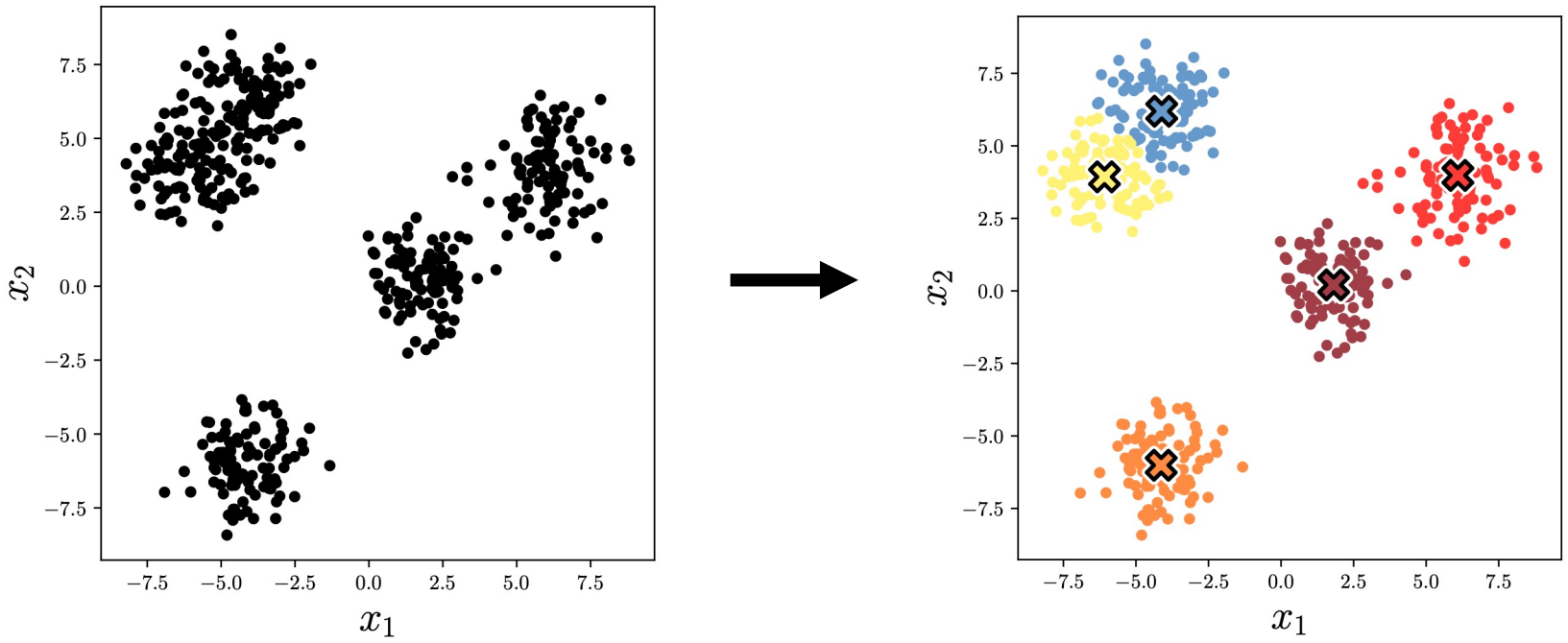


聚类源于压缩原则

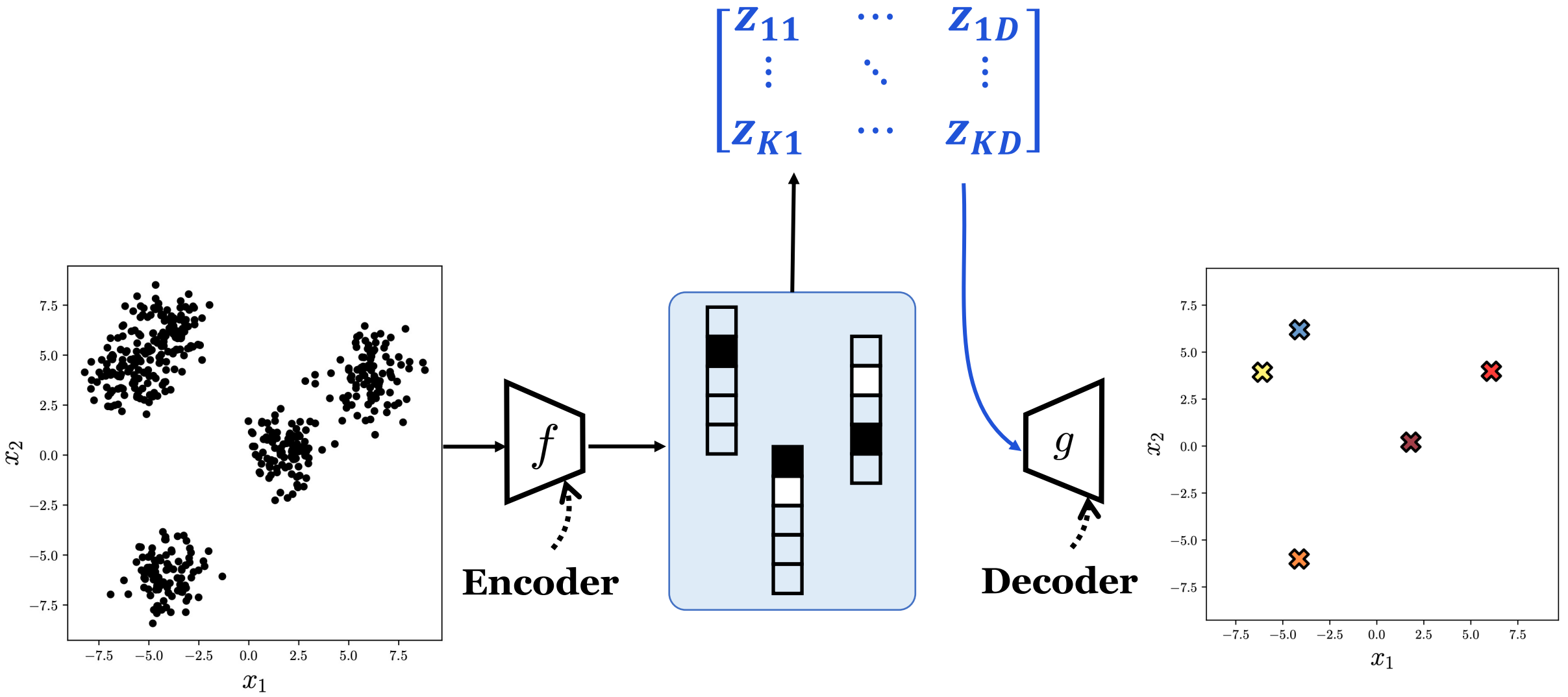
- 如果我们能仅用一个离散类别标签很好地概括一个信号，那么这个概括就可以作为进一步推理的更轻量级和更抽象的基础

k-means

- 将数据点映射到整数（即聚类）
- 映射方式：使得每个数据点尽可能接近其所属聚类的均值。



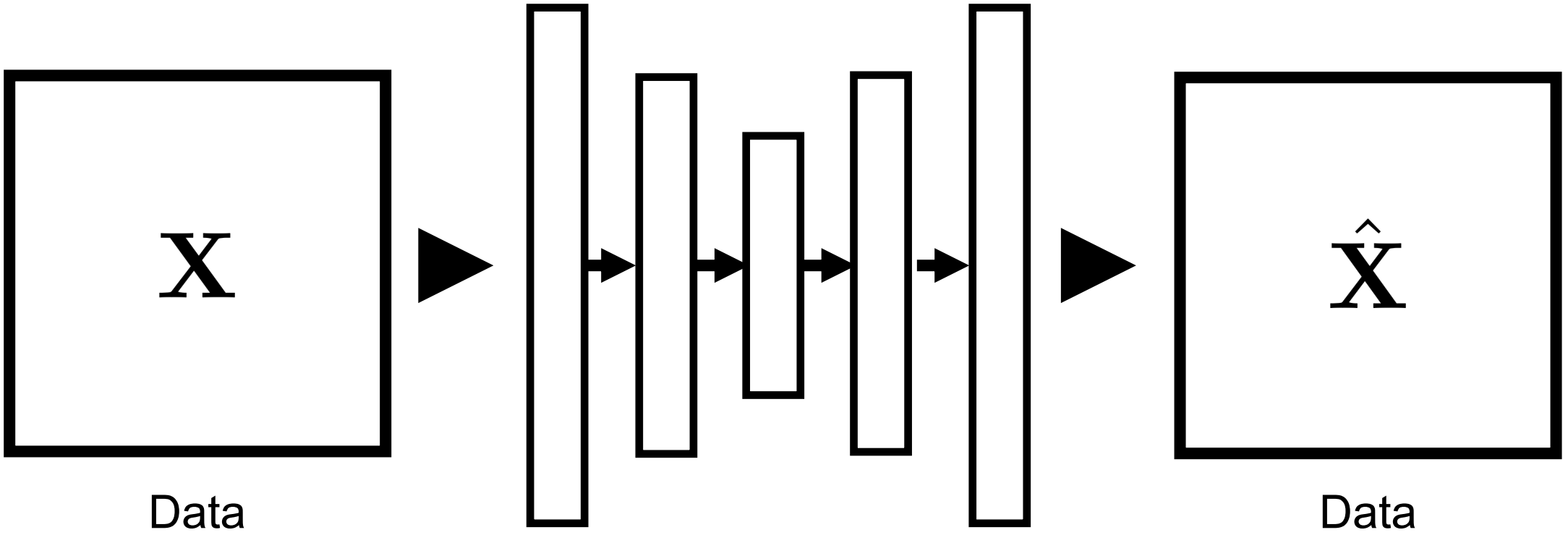
k-means


$\begin{bmatrix} z_{11} & \cdots & z_{1D} \\ \vdots & \ddots & \vdots \\ z_{K1} & \cdots & z_{KD} \end{bmatrix}$
f
Encoder
g
Decoder
x_1
x_2

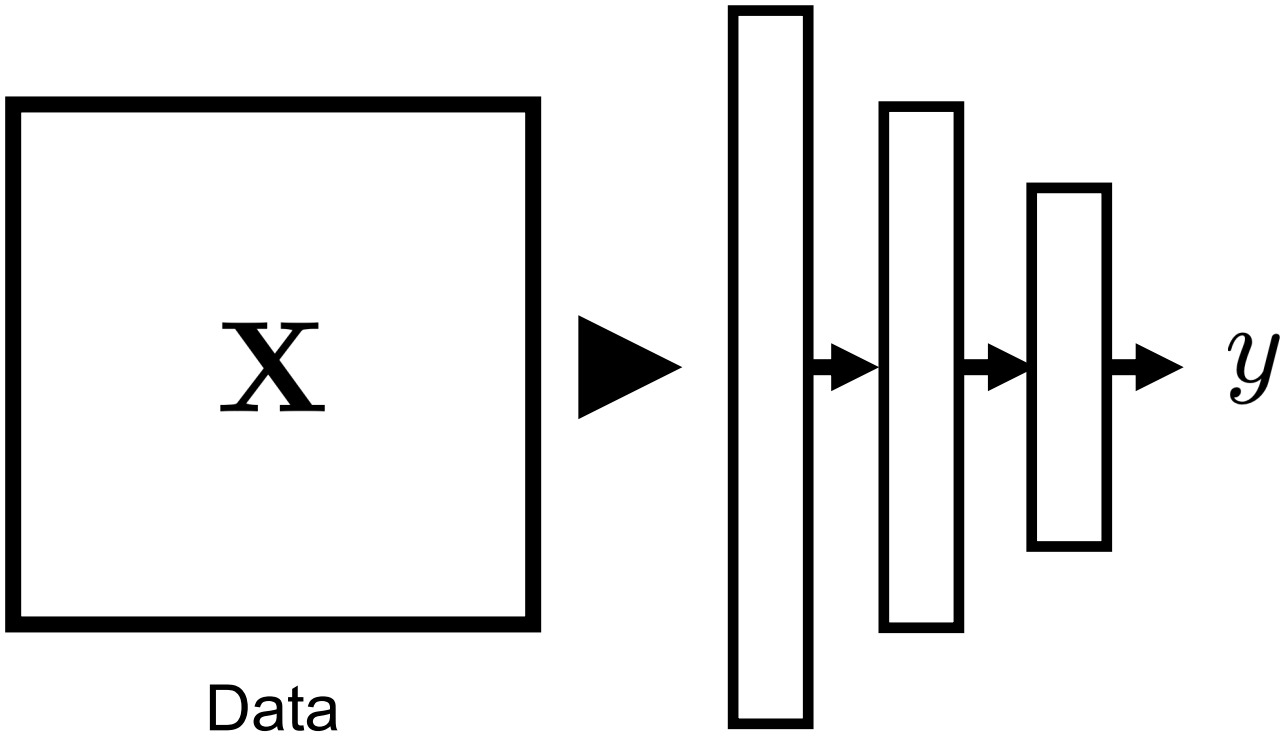
本章主要内容

- 表征学习问题
- 基于**压缩**的学习原则
  - 自编码器，对比学习，信息瓶颈，聚类
- 基于**预测**的学习原则
  - 预测编码，自监督学习，插补
- 信息论与**信息瓶颈**

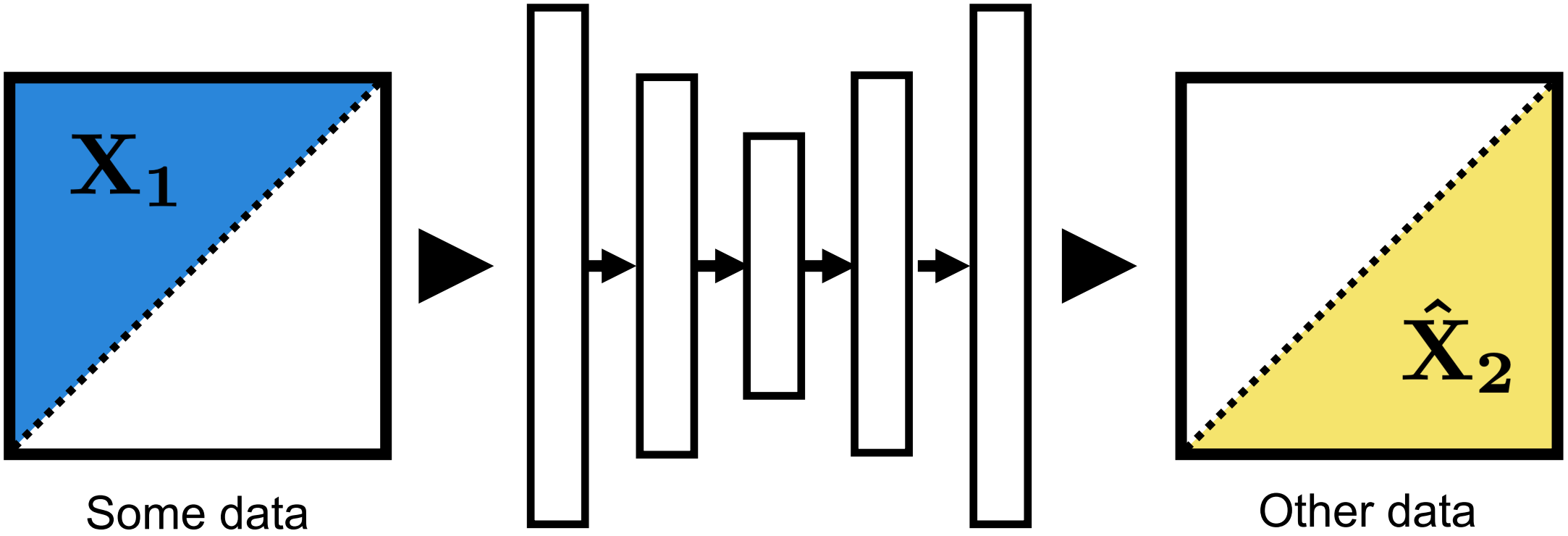
数据压缩



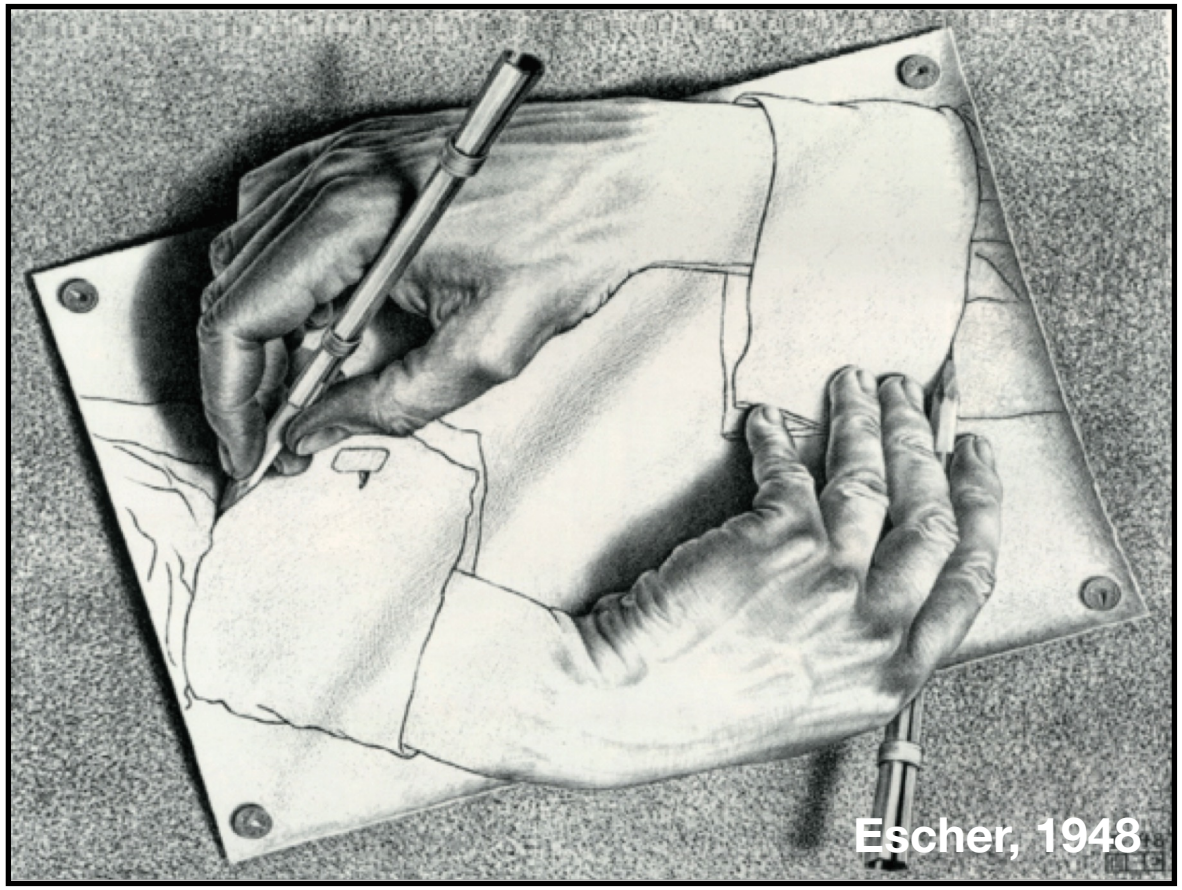
标签预测



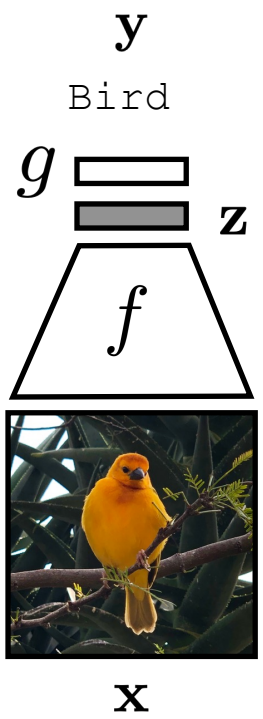
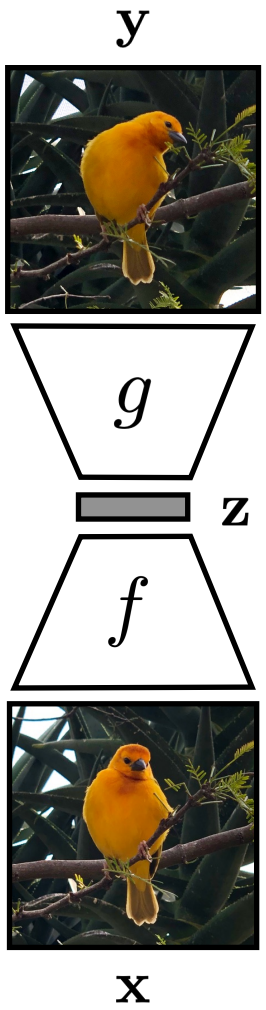
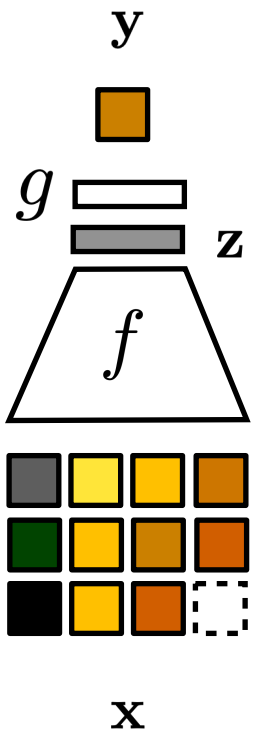
数据预测-自监督学习



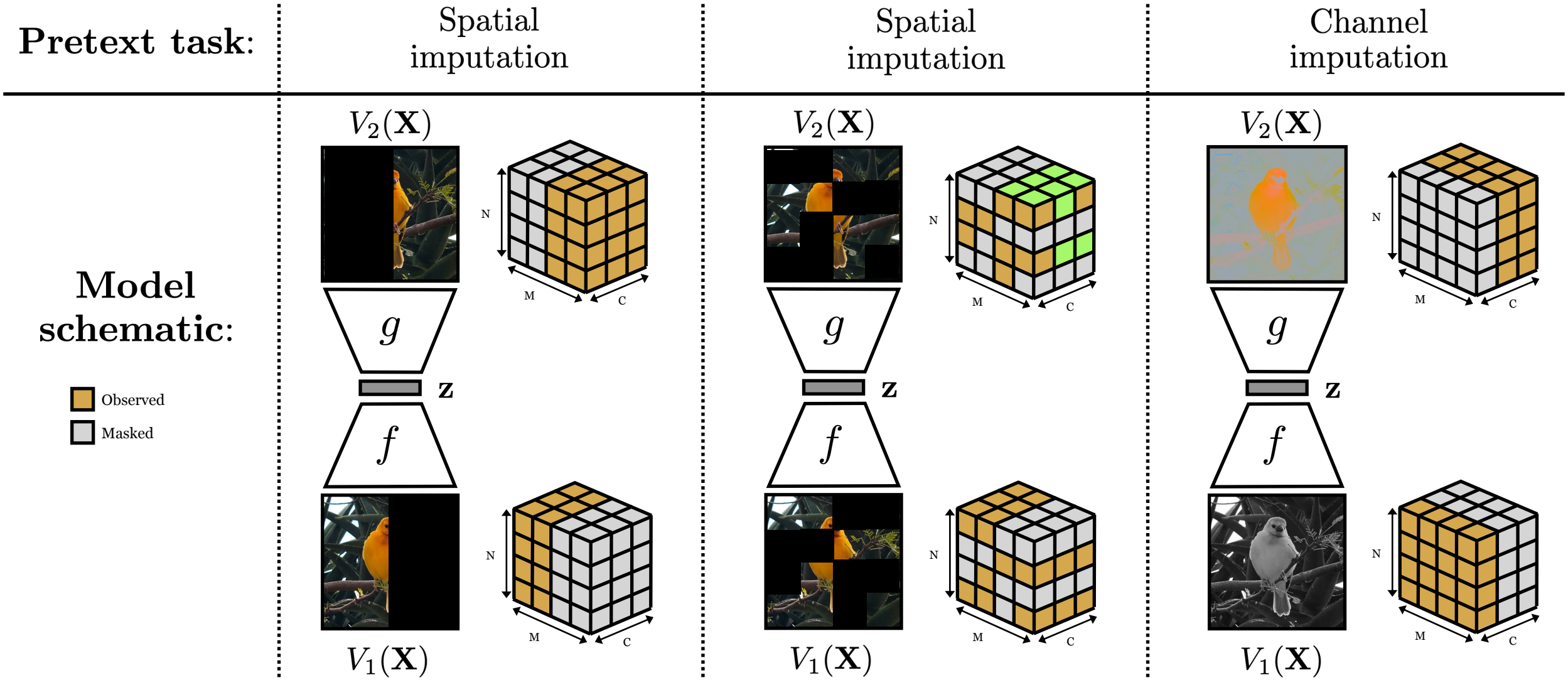
自监督学习



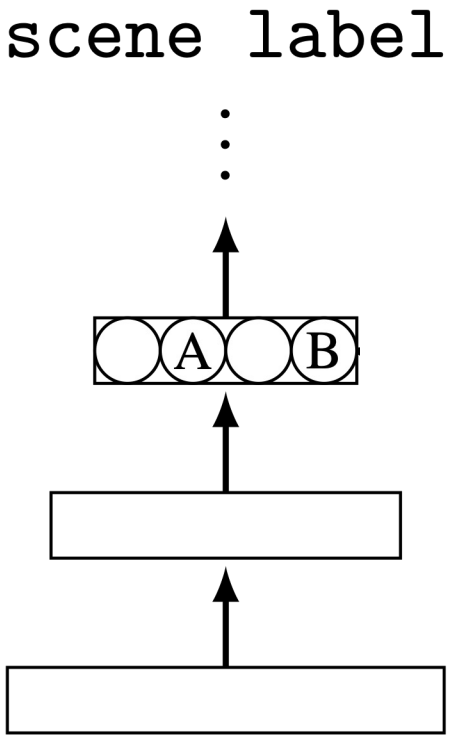
- 将**无监督**问题转化为**有监督**的经验风险最小化问题
- **代理任务（pretext task）**：直接从原始数据本身“炮制/构造”出标签（即预测目标）

Pretext task:	Class prediction	Future frame prediction	Next pixel prediction
Model schematic:			

缺失填补：统一的代理任务?

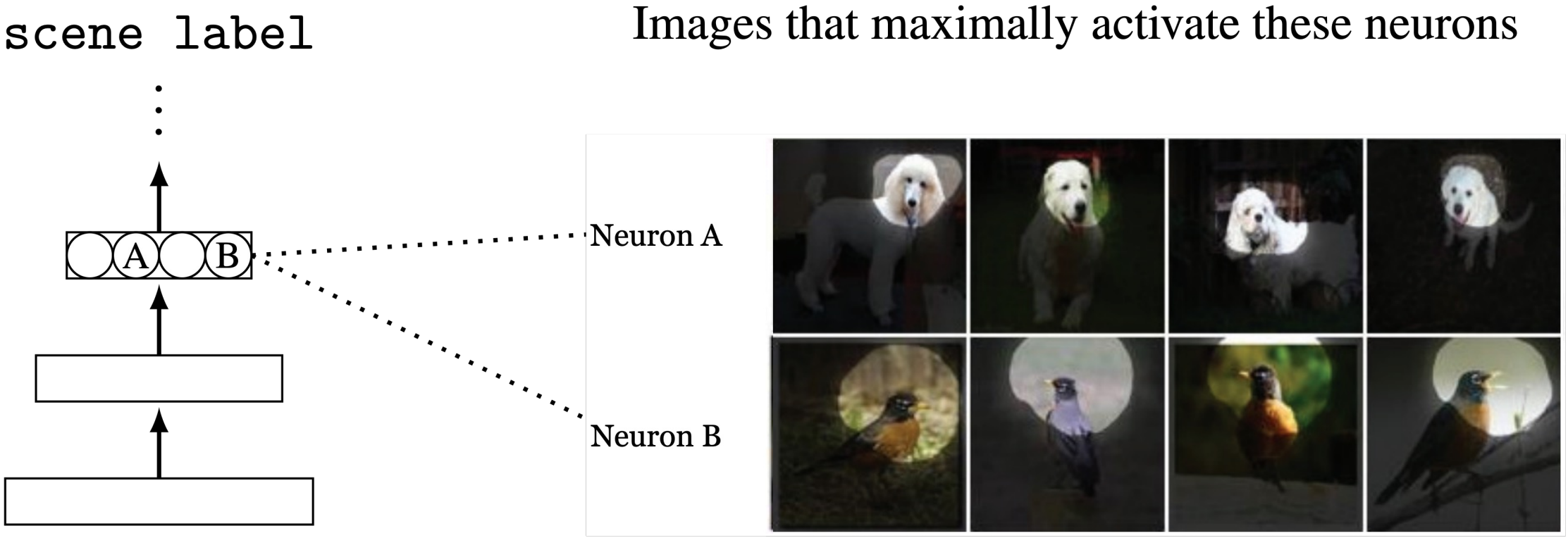


Object Detectors Emerge in Deep Scene CNNs



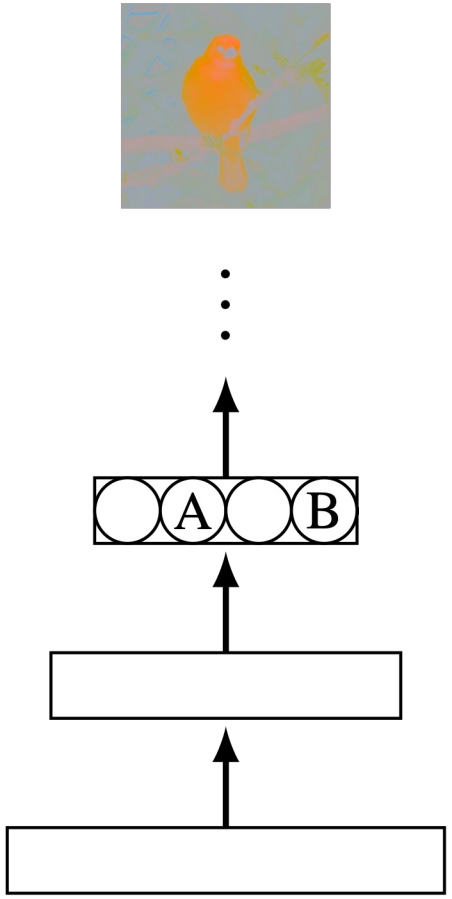
[Zhou, Khosla, Lapedriza, Oliva, Torralba 2015]

Object Detectors Emerge in Deep Scene CNNs



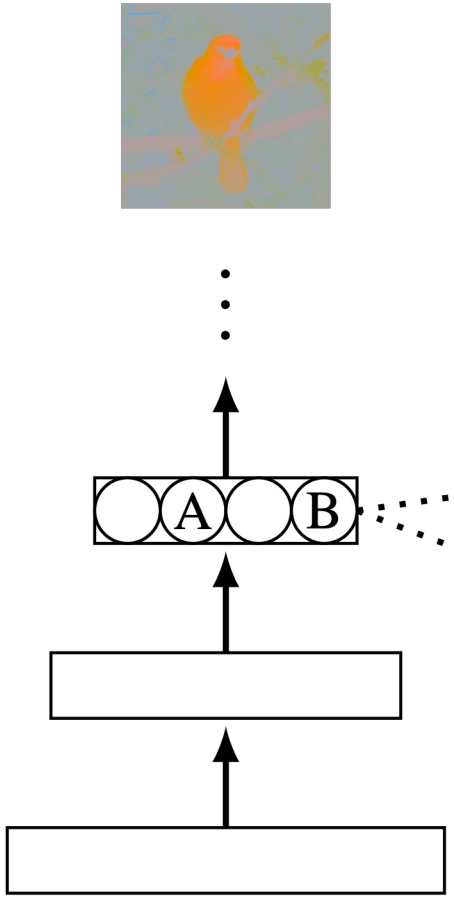
[Zhou, Khosla, Lapedriza, Oliva, Torralba 2015]

Object Detectors Emerge in Deep Scene CNNs



[Zhang, Isola, Efros 2016]

Object Detectors Emerge in Deep Scene CNNs



Images that maximally activate these neurons



[Zhang, Isola, Efros 2016]

本章主要内容

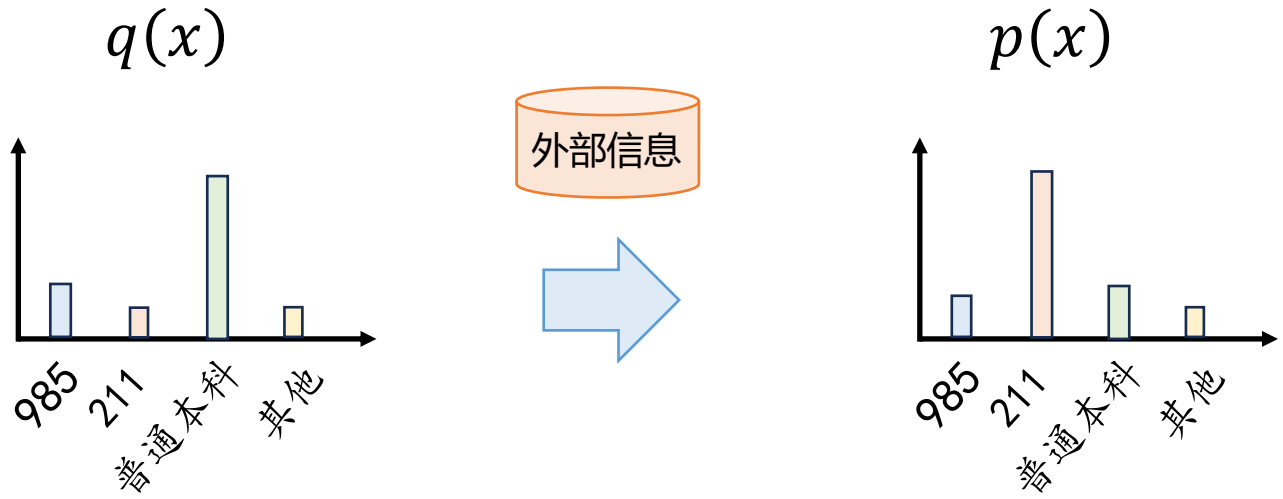
- 表征学习问题
- 基于**压缩**的学习原则
  - 自编码器，对比学习，信息瓶颈，聚类
- 基于**预测**的学习原则
  - 预测编码，自监督学习，插补
- 信息论与**信息瓶颈**

信息认知的改变

- 大学的类型：{985，211，普通本科，其他}

石河子大学是什么类型的大学？

认知的更新：$I(p||q)$



从$q(x)$到$p(x)$，我们的认知更新了多少？

认知更新 $I(p||q)$ 的性质

- **连续性**
- **非负性**
- **排列不变性**
  - $I\big(p(\text{“985”},\text{“211”}) \parallel q(\text{“985”},\text{“211”})\big) = I\big(p(\text{“211”},\text{“985”}) \parallel q(\text{“211”},\text{“985”})\big)$
- **对于均匀分布的单调性**
  - 从均匀分布到某个非均匀分布，应该有信息增益
- **满足自然链式法则**

➢ 若 $x = [x_L, x_R]$，$p(x) = p(x_L)P(x_R|x_L)$

$$I[p(\boldsymbol{x})\|q(\boldsymbol{x})] = I[p(\boldsymbol{x}_L)\|q(\boldsymbol{x}_L)] + \mathbb{E}_{p(\boldsymbol{x}_L)}\left[I[p(\boldsymbol{x}_R|\boldsymbol{x}_L)\|q(\boldsymbol{x}_R|\boldsymbol{x}_L)]\right]$$

KL满足$I(p||q)$的性质

- **KL散度**， 也叫相对熵或信息增益

$$D_{KL}(p \parallel q) \geq 0$$

离散情形：
$$D_{\mathbb{KL}}(p \parallel q) \triangleq \sum_{k=1}^{K} p_k \log \frac{p_k}{q_k}$$

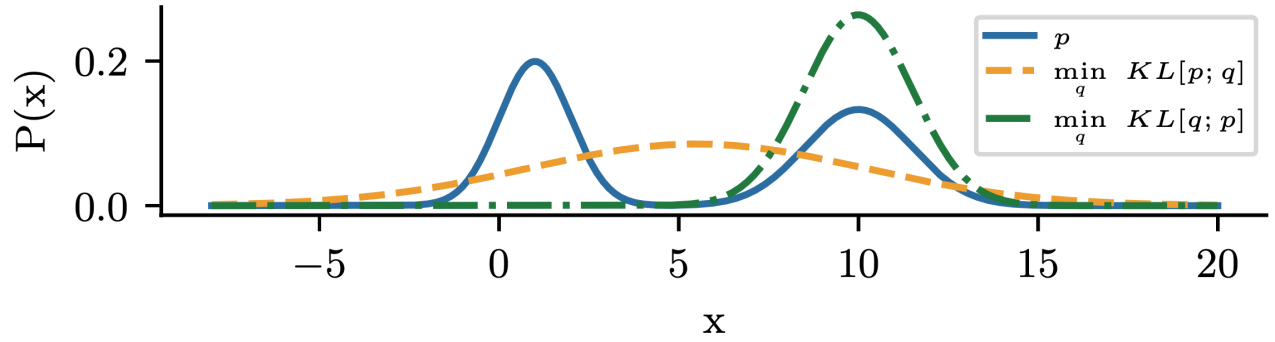
连续情形：
$$D_{\mathbb{KL}}(p \parallel q) \triangleq \int dx\, p(x) \log \frac{p(x)}{q(x)}$$

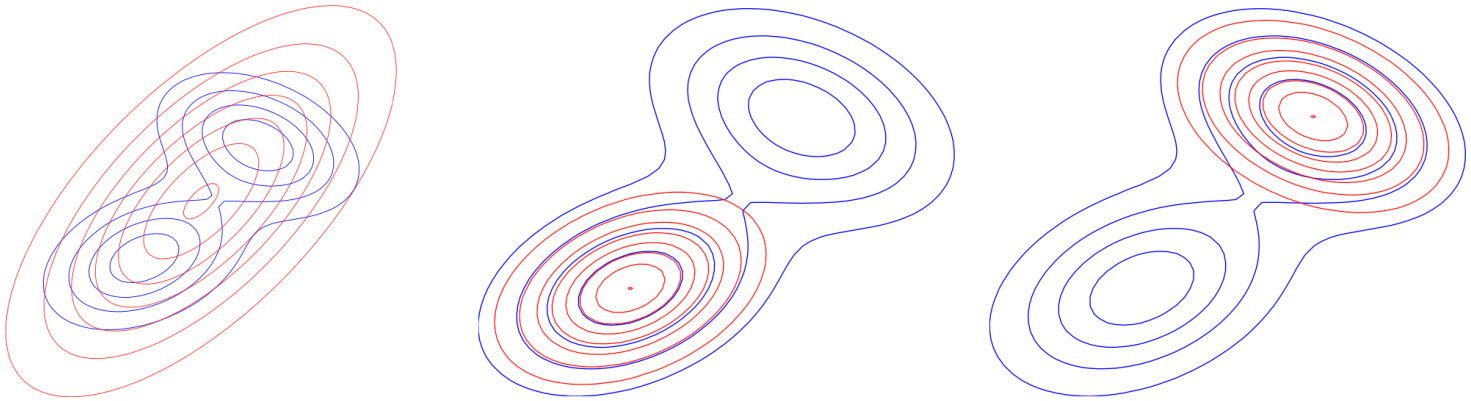
$$= \mathbb{E}_{p(x)} \left[log \frac{p(x)}{q(x)} \right]$$

$$= \int p(x) \log p(x)\, dx - p(x) \log q(x)\, dx$$

KL的非对称性

$$D_{KL}(p \parallel q) \neq D_{KL}(q \parallel p)$$



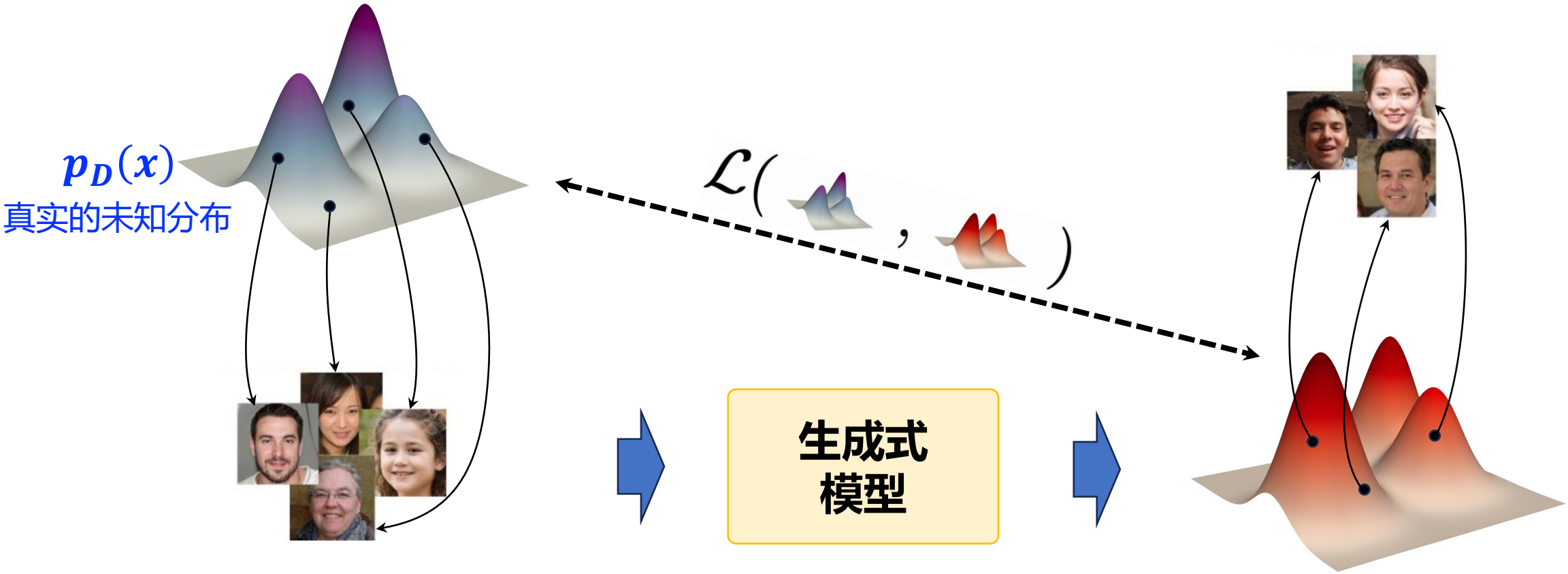


(a) (b) (c)

Min $D_{KL}(p \parallel q)$ Min $D_{KL}(q \parallel p)$

注：蓝色曲线为真实的分布p

生成式模型



$$p_E(x) = \frac{1}{N}\sum_{n=1}^{N}\delta(x_n - x)$$

经验分布

$$\min_{\theta} D_{KL}\left(p_E(x) \,\|\, p_\theta(x)\right)$$

$p_\theta(x)$

模型分布

计算 $D_{KL}\left(p_{\mathrm{E}}(x) \| p_{\theta}(x)\right)$

- 最小化KL散度，等价于极大似然

$$D_{KL}\left(p_E(x) \| p_\theta(x)\right) = \mathbb{E}_{p_E(x)}\left[\log \frac{p_E(x)}{p_\theta(x)}\right]$$

$$= \underbrace{\mathbb{E}_{p_{\mathrm{E}}(x)}[\log p_{\mathrm{E}}(x)]}_{\text{真实样本的熵取负号（常量）}} - \underbrace{\mathbb{E}_{p_{\mathrm{E}}(x)}[\log p_\theta(x)]}_{\text{对数似然}}$$

从$p_{\mathrm{E}}(x)$中采样，估计该期望

KL散度与交叉熵 — 判别式模型

$p(x,y) = p(x)p(y|x)$

- 训练集 $\{x_i, y_i\}_{i=1}^N$

$$D_{\mathbb{KL}}(p(x,y)||q(x,y)) = \iint dxdy\, p(x,y) \log \frac{p(x,y)}{q(x,y)} = \iint dxdy\, p(x,y) \left[\log \frac{p(x)}{q(x)} + \log \frac{p(y|x)}{q(y|x)}\right]$$

$$= \iint dxdy\, p(x,y) \log \frac{p(x)}{q(x)} + \iint dxdy\, p(x,y) \log \frac{p(y|x)}{q(y|x)}$$

$$= \underbrace{D_{\mathbb{KL}}(p(x)||q(x))}_{\text{边缘概率}} + \underbrace{\mathbb{E}_{p(x)}\left[D_{\mathbb{KL}}(p(y|x)||q(y|x))\right]}_{\text{条件概率}}$$

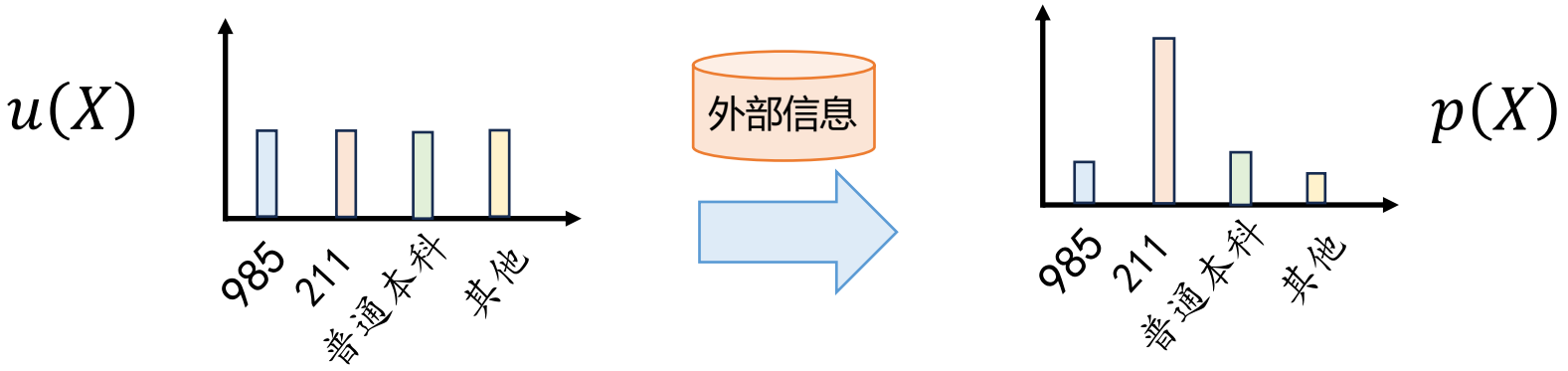
$$\mathbb{E}_{p(x)}[D_{\mathbb{KL}}\left(p(y|x) \parallel q(y|x)\right)] \triangleq \int dx\, p(x) \int dy\, p(y|x) \log \frac{p(y|x)}{q(y|x)}$$

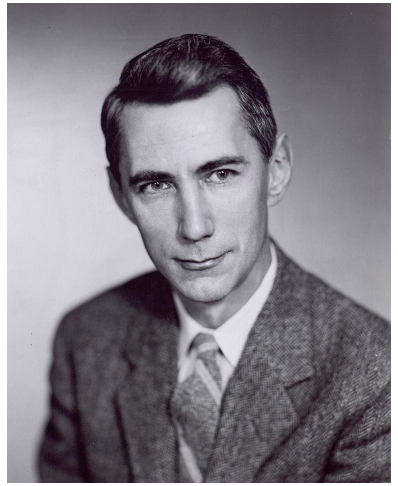
$$= \mathbb{E}_{\boldsymbol{x\sim p(x)}}\left[\underbrace{\mathbb{E}_{y\sim p(y|x)}[\log p(y|x)]}_{\text{常量}} - \underbrace{\mathbb{E}_{\boldsymbol{y\sim p(y|x)}}[\log q(y|x)]}_{\text{交叉熵}}\right]$$

判别式模型只对条件概率建模！假设 $\boldsymbol{D_{\mathbb{KL}}(p(x)||q(x)) = 0}$

信息熵

- 如果$q(X)$分布为均匀分布$u(X)$，意味着对该事件一无所知





克劳德·艾尔伍德·香农

$$D_{KL}(p||u) = \sum p(X)\log p(X) - \sum p(X)\log\frac{1}{K} = \sum p(X)\log p(X) + \log K$$

$$\boldsymbol{H(X)} = -\sum p(X)\log p(X) = \log K - D_{KL}(p||u) \leq \log K$$

消除不确定性的信息量

- 熵描述事物的不确定性；
- 熵只依赖于随机变量的分布，与随机变量的取值无关。

物理熵

“在一个封闭的孤立系统内，热量总是从高温流向低温，从有序走向无序，而且这一过程是不可逆转的”

—德国物理学家克劳修斯





物理熵

- 外部施加能量才能熵减



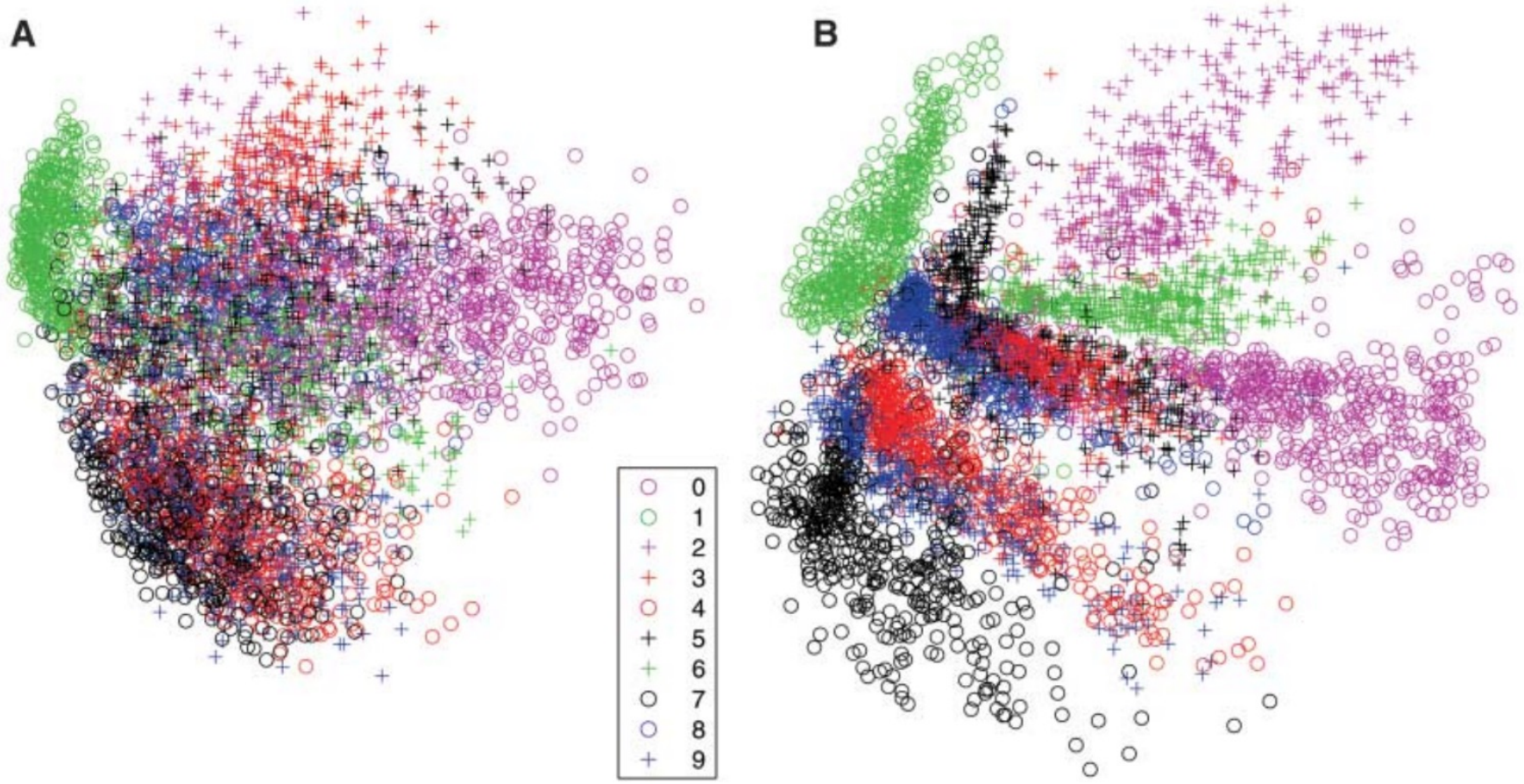
熵高



熵低

信息熵与机器学习

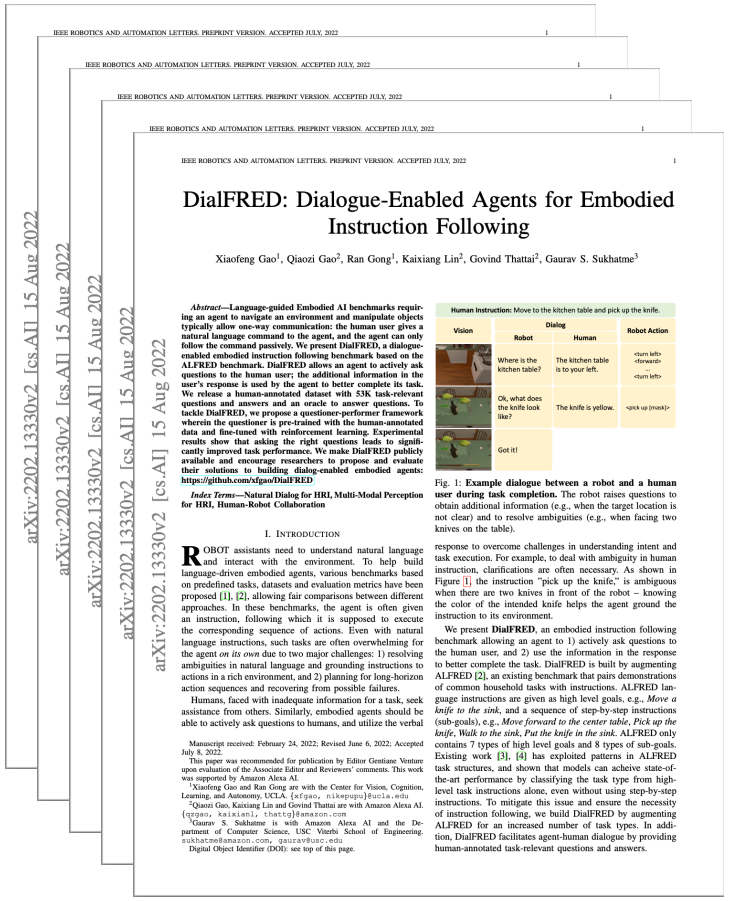
- 通过从数据进行模型学习来使熵降低



熵高

熵低

压缩编码



IEEE ROBOTICS AND AUTOMATION LETTERS. PREPRINT VERSION. ACCEPTED JULY, 2022 1

arXiv:2202.13330v2 [cs.AI] 15 Aug 2022

DialFRED: Dialogue-Enabled Agents for Embodied Instruction Following

Xiaofeng Gao[1], Qiaozi Gao[2], Ran Gong[1], Kaixiang Lin[2], Govind Thattai[2], Gaurav S. Sukhatme[3]

***Abstract*—Language-guided Embodied AI benchmarks requiring an agent to navigate an environment and manipulate objects typically allow one-way communication: the human user gives a natural language command to the agent, and the agent can only follow the command passively. We present DialFRED, a dialogue-enabled embodied instruction following benchmark based on the ALFRED benchmark. DialFRED allows an agent to actively ask questions to the human user; the additional information in the user's response is used by the agent to better complete its task. We release a human-annotated dataset with 53K task-relevant questions and answers and an oracle to answer questions. To tackle DialFRED, we propose a questioner-performer framework wherein the questioner is pre-trained with the human-annotated data and fine-tuned with reinforcement learning. Experimental results show that asking the right questions leads to significantly improved task performance. We make DialFRED publicly available and encourage researchers to propose and evaluate their solutions to building dialog-enabled embodied agents: https://github.com/xfgao/DialFRED**

***Index Terms*—Natural Dialog for HRI, Multi-Modal Perception for HRI, Human-Robot Collaboration**

I. Introduction

ROBOT assistants need to understand natural language and interact with the environment. To help build language-driven embodied agents, various benchmarks based on predefined tasks, datasets and evaluation metrics have been proposed [1], [2], allowing fair comparisons between different approaches. In these benchmarks, the agent is often given an instruction, following which it is supposed to execute the corresponding sequence of actions. Even with natural language instructions, such tasks are often overwhelming for the agent *on its own* due to two major challenges: 1) resolving ambiguities in natural language and grounding instructions to actions in a rich environment, and 2) planning for long-horizon action sequences and recovering from possible failures.

Humans, faced with inadequate information for a task, seek assistance from others. Similarly, embodied agents should be able to actively ask questions to humans, and utilize the verbal response to overcome challenges in understanding intent and task execution. For example, to deal with ambiguity in human instruction, clarifications are often necessary. As shown in Figure 1, the instruction "pick up the knife," is ambiguous when there are two knives in front of the robot – knowing the color of the intended knife helps the agent ground the instruction to its environment.

Manuscript received: February 24, 2022; Revised June 6, 2022; Accepted July 8, 2022.

This paper was recommended for publication by Editor Gentiane Venture upon evaluation of the Associate Editor and Reviewers' comments. This work was supported by Amazon Alexa AI.

[1]Xiaofeng Gao and Ran Gong are with the Center for Vision, Cognition, Learning, and Autonomy, UCLA. {xfgao, nikepupu}@ucla.edu

[2]Qiaozi Gao, Kaixiang Lin and Govind Thattai are with Amazon Alexa AI. {qzgao, kaixiani, thattg}@amazon.com

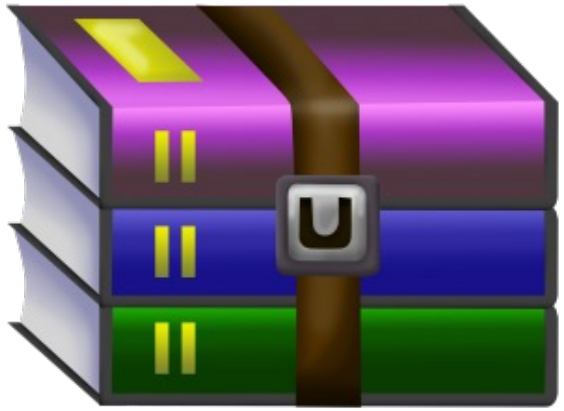
[3]Gaurav S. Sukhatme is with Amazon Alexa AI and the Department of Computer Science, USC Viterbi School of Engineering. sukhatme@amazon.com, gaurav@usc.edu

Digital Object Identifier (DOI): see top of this page.

Fig. 1: **Example dialogue between a robot and a human user during task completion.** The robot raises questions to obtain additional information (e.g., when the target location is not clear) and to resolve ambiguities (e.g., when facing two knives on the table).

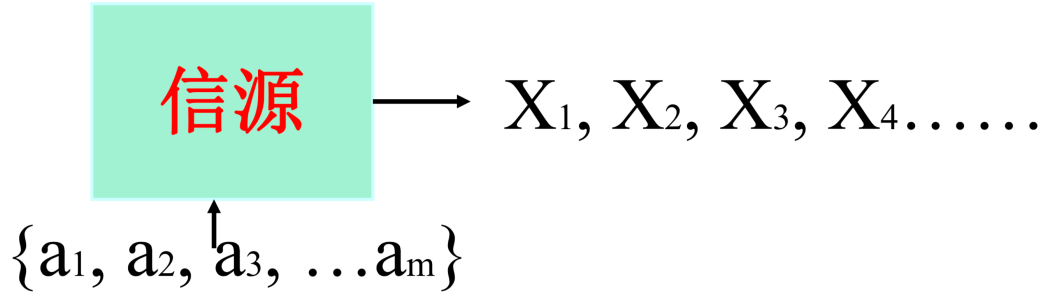
We present **DialFRED**, an embodied instruction following benchmark allowing an agent to 1) actively ask questions to the human user, and 2) use the information in the response to better complete the task. DialFRED is built by augmenting ALFRED [2], an existing benchmark that pairs demonstrations of common household tasks with instructions. ALFRED language instructions are given as high level goals, e.g., *Move a knife to the sink*, and a sequence of step-by-step instructions (sub-goals), e.g., *Move forward to the center table*, *Pick up the knife*, *Walk to the sink*, *Put the knife in the sink*. ALFRED only contains 7 types of high level goals and 8 types of sub-goals. Existing work [3], [4] has exploited patterns in ALFRED task structures, and shown that models can acheive state-of-the-art performance by classifying the task type from high-level task instructions alone, even without using step-by-step instructions. To mitigate this issue and ensure the necessity of instruction following, we build DialFRED by augmenting ALFRED for an increased number of task types. In addition, DialFRED facilitates agent-human dialogue by providing human-annotated task-relevant questions and answers.

- 编码长度
- 信息损失



无损压缩（信源编码）

- 信源的模型化 — 离散信源



- 如果随机序列中各个变量具有相同的概率分布，则称为离散平稳信源。
- 如果离散平稳信源的输出序列中各个变量是相互独立的，即前一个符号的出现不影响以后任何一个符号出现的概率，则称为离散无记忆平稳信源，否则称为离散有记忆平稳信源。

无损压缩（信源编码）

- 信源的模型化—离散信源
  - 假定信源的符号系列为$\boldsymbol{A=\{a_1,a_2,\ldots,a_j\}}$，信源产生符号$\boldsymbol{a_j}$的事件概率是$\boldsymbol{P(a_j)}$，并有:

$$\sum_{i=1}^{j} P(a_i)=1$$

- 一个$J\times1$向量$\boldsymbol{z=[P(a_1),P(a_2),\ldots,P(a_m)]^T}$，用于表示所有信源符号的概率集合。

- 有限总体集合$(\boldsymbol{A},\boldsymbol{z})$完全描述了信息源。

离散无记忆信源的编码

- 无记忆：不考虑字符之间的相关性
- 如果对字符a_j的编码长度为L_j，则X的平均码长为:

$$L = \sum_{j=1}^{m} p_j * L_j$$

- 平均码长的下界为$H(X)$

$$L = \sum_{j=1}^{m} p_j * L_j \geq H(X) = -\sum_{j=1}^{m} p_j * \log_2 p_j$$ （证明略）

  - 当 $L_j = -\log_2 p_j$ 时，等号成立
  - 对出现概率较大的符号取较短的码长，对出现概率较小的符号取较大的码长
  - 压缩的基本途径之一：使各字符的编码长度尽量等于字符的信息量

概率模型对编码的影响

符号	a_1	a_2	……	a_m	平均码长
真实概率	p_1	p_2	……	p_m	
理想编码长度	$-\log_2 p_1$	$-\log_2 p_2$	……	$-\log_2 p_m$	$-\sum p_j \log p_j = H(p)$
建模概率	q_1	q_2	……	q_m	
实际编码长度	$-\log_2 q_1$	$-\log_2 q_2$	……	$-\log_2 q_m$	$-\sum p_j \log q_j = H_{ce}(p,q)$

熵

交叉熵

- 交叉熵是按照分布q的最优编码对真实分布p的信息进行编码的平均长度
- 实际编码长度与理想编码长度之差：

$$H_{ce}(p,q) - H(p) = D_{KL}(p||q)$$

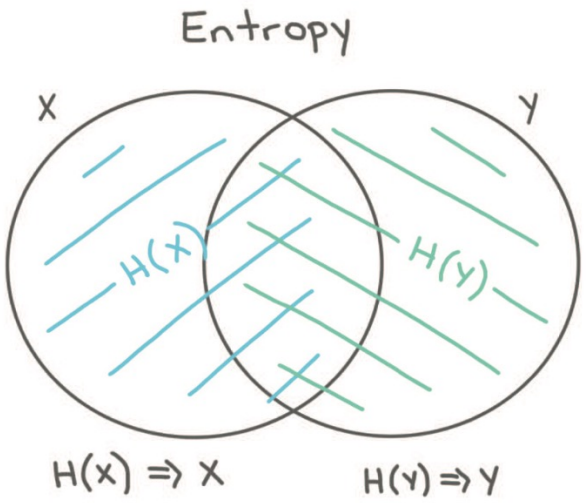
互信息

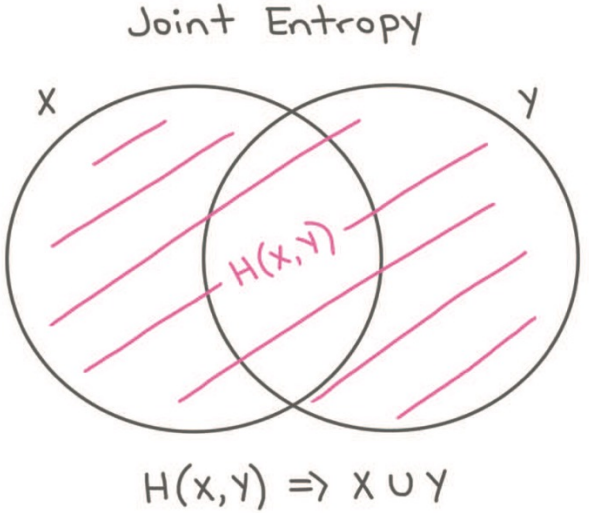
- 衡量两个随机变量之间的依赖性
- 从$p(X)p(Y)$到$p(X,Y)$的信息增益

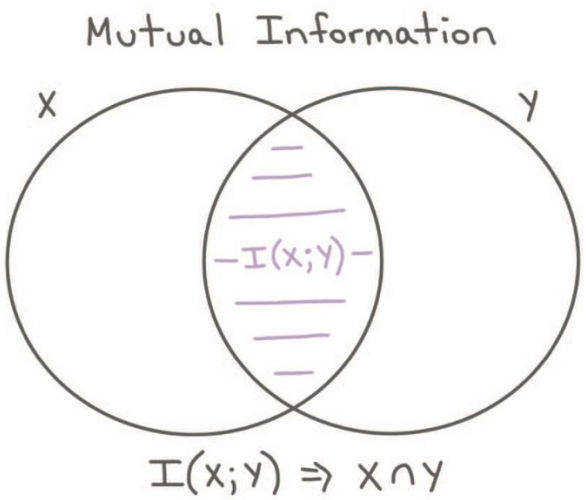
$$\mathbb{I}(X;Y) \triangleq D_{\mathbb{KL}}(p(x,y) \parallel p(x)p(y)) = \sum_{y\in Y}\sum_{x\in X} p(x,y)\log\frac{p(x,y)}{p(x)\,p(y)}$$

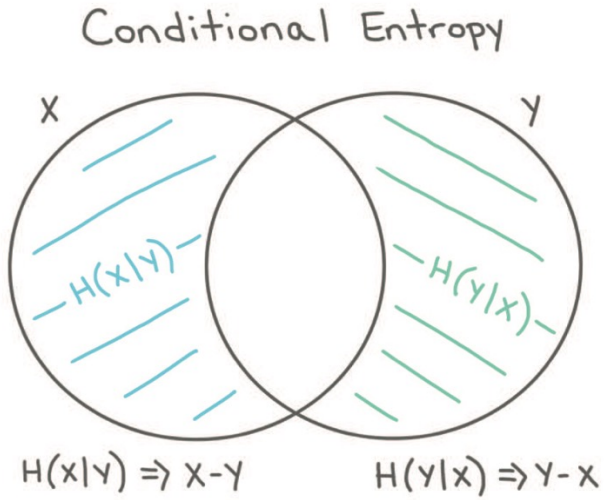
$$\mathbb{I}(X;Y) = \mathbb{H}(X) - \mathbb{H}(X|Y) = \mathbb{H}(Y) - \mathbb{H}(Y|X)$$

互信息


Entropy
X
Y
H(X)
H(Y)
H(X) ⇒ X
H(Y) ⇒ Y

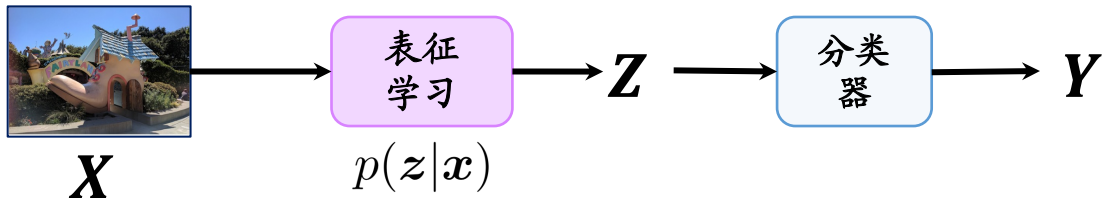

Joint Entropy
X
Y
H(X,Y)
H(X,Y) ⇒ X ∪ Y


Mutual Information
X
Y
I(X;Y)
I(X;Y) ⇒ X ∩ Y


Conditional Entropy
X
Y
H(X|Y)
H(Y|X)
H(X|Y) ⇒ X-Y
H(Y|X) ⇒ Y-X

信息瓶颈—判别式模型

- 从输入X中学习表征Z，用于预测标签Y



- 若 $y \perp x|z$，则称Z对于预测Y是**充分的**
  - Z包含了X中的所有关于Y的信息。

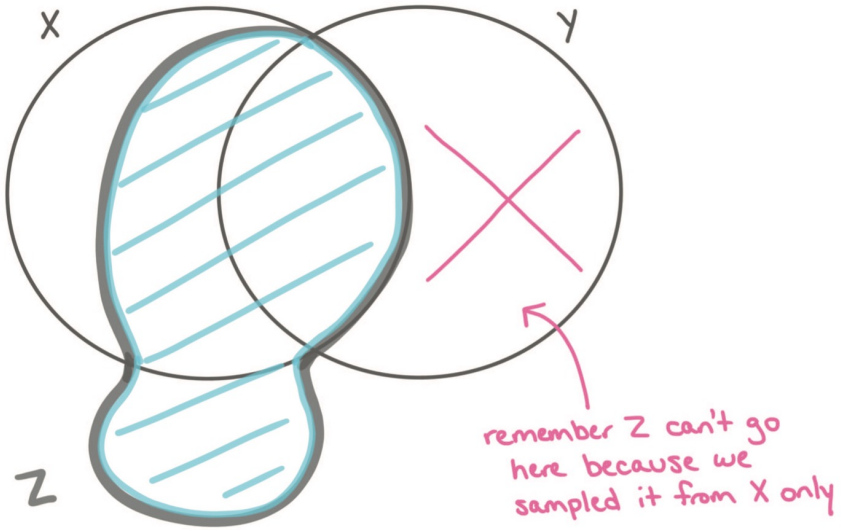
$$\mathbb{H}(y|z) = \mathbb{H}(y|x) \iff \mathbb{I}(z;y) = \mathbb{I}(x;y)$$

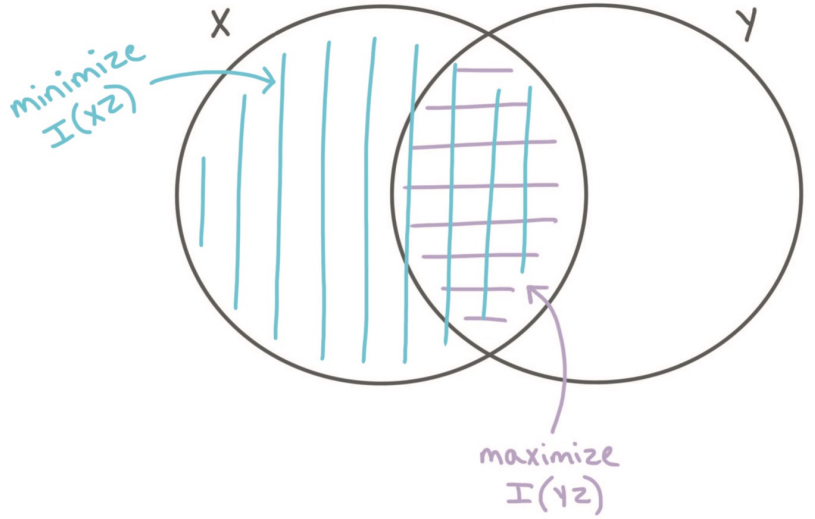
$$\mathbb{I}(z;y) = \mathbb{H}(y) - \mathbb{H}(y|z)$$
$$\mathbb{I}(x;y) = \mathbb{H}(y) - \mathbb{H}(y|z)$$

信息瓶颈原则

- 若Z对于预测Y是充分的，且找不到其他的Z，使$\mathbb{I}(z;x)$具有更小的值，则称Z是**最小充分统计的**
- **优化目标：**找到一个Z，使得$\mathbb{I}(z;y)$最大，而$\mathbb{I}(z;x)$最小

$$\boldsymbol{min\ \beta\ \mathbb{I}(z;x) - \mathbb{I}(z;y)}$$





西南交通大学
Southwest Jiaotong University

SCAI011112 视觉计算

Thank You